US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12417913
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Kamikawa; Takeshi et al.

Method for flattening a surface on an epitaxial lateral growth layer

Abstract

A method for flattening a surface on an epitaxial lateral overgrowth (ELO) layer, resulting in obtaining a smooth surface with island-like III-nitride semiconductor layers. The island-like III-nitride semiconductor layers are formed by stopping the growth of the ELO layers before they coalesce to each other. Then, a growth restrict mask is removed before at least some III-nitride device layers are grown. Removing the mask decreases an excess gases supply to side facets of the island-like III-nitride semiconductor layers, which can help to obtain a smooth surface on the island-like III-nitride semiconductor layers. The method also avoids compensation of a p-type layer by decomposed n-type dopant from the mask, such as Silicon and Oxygen atoms.

Inventors:	**Kamikawa; Takeshi (Santa Barbara, CA), Gandrothula; Srinivas (Santa Barbara, CA)**
Applicant:	**The Regents of the University of California** (Oakland, CA)
Family ID:	**1000008820868**
Assignee:	**THE REGENTS OF THE UNIVERSITY OF CALIFORNIA (Oakland, CA)**
Appl. No.:	**17/429570**
Filed (or PCT Filed):	**March 02, 2020**
PCT No.:	**PCT/US2020/020647**
PCT Pub. No.:	**WO2020/180785**
PCT Pub. Date:	September 10, 2020

Prior Publication Data

Document Identifier	**Publication Date**
US 20220108883 A1	Apr. 07, 2022

Related U.S. Application Data

us-provisional-application US 62812453 20190301

Publication Classification

Int. Cl.: **H01L21/02** (20060101); **H10D62/85** (20250101)

U.S. Cl.:

CPC **H01L21/0254** (20130101); **H01L21/02458** (20130101); **H01L21/02639** (20130101); **H10D62/8503** (20250101); H01L21/02387 (20130101); H01L21/02455 (20130101); H01L21/02538 (20130101)

Field of Classification Search

CPC: C01B (21/072-0728); C01B (21/064-0648); C01B (21/0632)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6265289	12/2000	Zheleva	257/190	H01L 21/0254
2003/0143771	12/2002	Kidoguchi	257/E33.068	H10H 20/835
2006/0266281	12/2005	Beaumont et al.	N/A	N/A
2007/0164306	12/2006	Nakahata	257/E33.033	H01L 21/02389
2009/0258453	12/2008	Lee et al.	N/A	N/A
2010/0001289	12/2009	Frayssinet et al.	N/A	N/A
2015/0137137	12/2014	Chowdhury et al.	N/A	N/A
2015/0179875	12/2014	Jang	438/492	H01L 21/0254
2018/0190789	12/2017	Bour et al.	N/A	N/A
2022/0123166	12/2021	Kamikawa et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1518138	12/2003	CN	N/A
103608923	12/2013	CN	N/A
104733365	12/2014	CN	N/A
113439322	12/2020	CN	N/A
113439322	12/2020	CN	N/A
1748470	12/2010	EP	N/A
2000277437	12/1999	JP	N/A
2001520169	12/2000	JP	N/A

2002009004	12/2001	JP	N/A
2002299251	12/2001	JP	N/A
2006316307	12/2005	JP	N/A
2006316307	12/2005	JP	N/A
2007001857	12/2006	JP	N/A
2007184504	12/2011	JP	N/A
2014520405	12/2013	JP	N/A
2015500573	12/2014	JP	N/A
2013/089459	12/2012	WO	N/A
2019055936	12/2018	WO	N/A
2019222669	12/2018	WO	N/A
2020150511	12/2019	WO	N/A

OTHER PUBLICATIONS

Chinese First Office Action (with English translation) dated Aug. 30, 2023 for Chinese Patent Application No. 202080024824.6. cited by applicant
PCT International Search Report and Written Opinion dated May 12, 2020 for PCT Application No. PCT/US2020/020647. cited by applicant
Zytkiewicz, Zr. Epitaxial Lateral Overgrowth—A Tool for Dislocation Blockade in Multilayer Systems. Acta Physica Polonica A. 1998. vol. 94(2). p. 220, figure 1; p. 225, figure 2. cited by applicant
Extended European Search Report dated Apr. 4, 2022 for EP Application No. 20766879.9. cited by applicant
Zheleva TS et al: “Pendeo-epitaxy—a new approach for lateral growth of gallium nitride structures”, MRS Internet Journal of Nitride Semiconductor Research, Materials Research Society, Warrendale, PA, US, vol. 4SI, Nov. 30, 1999 (Nov. 30, 1999), 7 pages. cited by applicant
Japanese Final Office Action dated Aug. 23, 2023 for Japanese Application No. 2021-551540. cited by applicant
Chinese Office Action dated Aug. 30, 2023 for Chinese Application No. 202080024824.6. cited by applicant
Japanese Notice of Reasons for Rejection dated Nov. 25, 2022 for JP Application No. 2021-551540. cited by applicant
Chinese Second Office Action (with English translation) dated May 25, 2024 for Chinese Patent Application No. 202080024824.6. cited by applicant
Chinese Third Office Action (with English translation) dated Nov. 16, 2024 for Chinese Patent Application No. 202080024824.6. cited by applicant

---

Primary Examiner: Landau; Matthew C

Assistant Examiner: Nix; Nora T.

Attorney, Agent or Firm: Gates & Cooper LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit under 35 U.S.C. Section 119(e) of the following co-pending and commonly-assigned application: (2) U.S. Provisional Application Ser. No. 62/812,453, filed on Mar. 1, 2019, by Takeshi Kamikawa and Srinivas Gandrothula; entitled “METHOD FOR FLATTENING A SURFACE ON AN

EPITAXIAL LATERAL GROWTH LAYER,"; which application is incorporated by reference herein. (3) This application is related to the following co-pending and commonly-assigned applications: (4) U.S. Utility patent application Ser. No. 16/608,071; filed on Oct. 24, 2019, by Takeshi Kamikawa, Srinivas Gandrothula, Hongjian Li and Daniel A. Cohen, entitled "METHOD OF REMOVING A SUBSTRATE," which application claims the benefit under 35 U.S.C. Section 365(c) of co-pending and commonly-assigned PCT International Patent Application No. PCT/US18/31393, filed on May 7, 2018, by Takeshi Kamikawa, Srinivas Gandrothula, Hongjian Li and Daniel A. Cohen, entitled "METHOD OF REMOVING A SUBSTRATE," which application claims the benefit under 35 U.S.C. Section 119(e) of co-pending and commonly-assigned U.S. Provisional Patent Application No. 62/502,205, filed on May 5, 2017, by Takeshi Kamikawa, Srinivas Gandrothula, Hongjian Li and Daniel A. Cohen, entitled "METHOD OF REMOVING A SUBSTRATE,"; (5) PCT International Patent Application No. PCT/US18/51375, filed on Sep. 17, 2018, by Takeshi Kamikawa, Srinivas Gandrothula and Hongjian Li, entitled "METHOD OF REMOVING A SUBSTRATE WITH A CLEAVING TECHNIQUE," which application claims the benefit under 35 U.S.C. Section 119(e) of co-pending and commonly-assigned U.S. Provisional Patent Application No. 62/559,378, filed on Sep. 15, 2017, by Takeshi Kamikawa, Srinivas Gandrothula and Hongjian Li, entitled "METHOD OF REMOVING A SUBSTRATE WITH A CLEAVING TECHNIQUE,"; (6) PCT International Patent Application No. PCT/US19/25187, filed on Apr. 1, 2019, by Takeshi Kamikawa, Srinivas Gandrothula and Hongjian Li, entitled "METHOD OF FABRICATING NONPOLAR AND SEMIPOLAR DEVICES USING EPITAXIAL LATERAL OVERGROWTH," which application claims the benefit under 35 U.S.C. Section 119(e) of co-pending and commonly-assigned U.S. Provisional Patent Application Ser. No. 62/650,487, filed on Mar. 30, 2018, by Takeshi Kamikawa, Srinivas Gandrothula, and Hongjian Li, entitled "METHOD OF FABRICATING NONPOLAR AND SEMIPOLAR DEVICES BY USING LATERAL OVERGROWTH,"; (7) PCT International Patent Application No. PCT/US19/32936, filed on May 17, 2019, by Takeshi Kamikawa and Srinivas Gandrothula, entitled "METHOD FOR DIVIDING A BAR OF ONE OR MORE DEVICES," which application claims the benefit under 35 U.S.C. Section 119(e) of co-pending and commonly-assigned U.S. Provisional Application Ser. No. 62/672,913, filed on May 17, 2018, by Takeshi Kamikawa and Srinivas Gandrothula, entitled "METHOD FOR DIVIDING A BAR OF ONE OR MORE DEVICES,"; (8) PCT International Patent Application No. PCT/US19/34686, filed on May 30, 2019, by Srinivas Gandrothula and Takeshi Kamikawa, entitled "METHOD OF REMOVING SEMICONDUCTING LAYERS FROM A SEMICONDUCTING SUBSTRATE," which application claims the benefit under 35 U.S.C. Section 119(e) of co-pending and commonly-assigned U.S. Provisional Application Ser. No. 62/677,833, filed on May 30, 2018, by Srinivas Gandrothula and Takeshi Kamikawa, entitled "METHOD OF REMOVING SEMICONDUCTING LAYERS FROM A SEMICONDUCTING SUBSTRATE,"; (9) PCT International Patent Application No. PCT/US19/59086, filed on Oct. 31, 2019, by Takeshi Kamikawa and Srinivas Gandrothula, entitled "METHOD OF OBTAINING A SMOOTH SURFACE WITH EPITAXIAL LATERAL OVERGROWTH," which application claims the benefit under 35 U.S.C. Section 119(e) of co-pending and commonly-assigned U.S. Provisional Application Ser. No. 62/753,225, filed on Oct. 31, 2018, by Takeshi Kamikawa and Srinivas Gandrothula, entitled "METHOD OF OBTAINING A SMOOTH SURFACE WITH EPITAXIAL LATERAL OVERGROWTH,"; and (10) PCT International Patent Application No. PCT/US20/13934, filed on Jan. 16, 2020, by Takeshi Kamikawa, Srinivas Gandrothula and Masahiro Araki, entitled "METHOD FOR REMOVAL OF DEVICES USING A TRENCH," which application claims the benefit under 35 U.S.C. Section 119(e) of co-pending and commonly-assigned U.S. Provisional Application Ser. No. 62/793,253, filed on Jan. 16, 2019, by Takeshi Kamikawa, Srinivas Gandrothula and Masahiro Araki, entitled "METHOD FOR REMOVAL OF DEVICES USING A TRENCH,"; all of which applications are incorporated by reference herein.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) This invention relates to a method fix flattening a surface on an epitaxial lateral growth layer.

2. Description of the Related Art

(2) Currently, some device manufacturers use III-nitride-based substrates, such as GaN or MN substrates, to produce laser diodes (LDs) and light-emitting diodes (LEDs) for lighting, optical storage, etc. However, it is well known that III-nitride-based substrates are very expensive.

(3) Prior attempts have been made to remove III-nitride-based semiconductor layers from a III-nitride-based substrate to recycle the substrate. In one example, III-nitride layers are first grown on the substrate using epitaxial lateral overgrowth (ELO) with a growth restrict mask. The growth of the III-nitride layers is stopped before the III-nitride layers coalesce with each other. This makes it easy to remove the resulting island-like III-nitride semiconductor layers, after a device process is performed that makes a ridge structure on the island-like III-nitride semiconductor layers.

(4) Using this method allows one to grow III-nitride device layers on the discrete ELO III-nitride layers using metal organic chemical vapor deposition (MOCVD). In conventional epitaxial growth using MOCVD, growth can be performed on a flat substrate or a flat substrate with a template layer. However, the III-nitride device layers are grown on discrete epilayers, and such growth does not obtain a flat surface epi-layer on a substrate with discrete ELO III-nitride layers.

(5) Deteriorating surface roughness means each layer has an in-plane distribution of a thickness. To improve the characteristics of the device, the in-plane distribution of the thickness of each layer needs to decrease. For example, if the in-plane distribution of the thickness is in the p-layer, the optical confinement factor is different in each device. Improving surface roughness decreases the in-plane distribution of the thickness.

(6) In another aspect, the growth restrict mask typically is SiO.sub.2, SiN, etc. In these cases, Si and O are the n-type dopant for a GaN layer. It may be possible to compensate the n-type layers while p-type layers are growing, if the growth restrict mask is exposed or not covered by something.

(7) The purpose of the present invention is to obtain a smooth surface after the growth of the III-nitride device layer.

SUMMARY OF THE INVENTION

(8) To overcome the limitations in the prior art described above, and to overcome other limitations that will become apparent upon reading and understanding this specification, the present invention discloses a method for obtaining a smooth surface with a discrete ELO III-nitride layer. Moreover, the present invention has succeeded in improving the surface roughness of an III-nitride device layer on the discrete ELO III-nitride layer.

(9) In another aspect, its critical for the sake of mounting devices with their top-side down to suppress an edge-growth at the edge of a bar of the device. Non-uniform supply of atoms from the material gases causes this phenomenon. The present invention has succeeded in suppressing the edge-growth.

(10) Specifically, this invention performs the following steps: ELO III-nitride layers are grown on a substrate using a growth restrict mask via MOCVD or other methods; the ELO III-nitride layers remain separate and discrete from each other; the substrate is removed from the MOCVD reactor to remove the growth restrict mask; the growth restrict mask is removed by wet or dry etching; III-nitride device layers are grown on the ELO II-nitride layers and substrate after removal of the growth restrict mask, resulting in island-like III-nitride semiconductor layers; devices are fabricated using the island-like III-nitride semiconductor layers; bars of the devices are removed from the substrate; and the bars of the devices are divided into chips or individual devices using a cleaving method.

(11) After growing the III-nitride device layers, it is difficult to flatten their surface. This is

especially true for III-nitride semiconductor layers that contain ternary and quaternary compound III-nitride semiconductor layers.
(12) Research has disclosed that unevenness in growth is caused by a non-uniform supply of gases, which affects the flatness of the surface and results in a non-uniform emitting pattern for the active layer. Moreover, the unevenness in growth happens at portions near the edges of the island-like III-nitride semiconductor layers, which is caused by a difference in the amount of supply atoms at each portion.
(13) In the present invention, this problem of the unevenness of growth is enhanced by the ELO III-nitride layers being separated, which increases the number of edges. Furthermore, since the width of a flat surface region is narrow as compared to conventional growth using a common wafer, the unevenness of growth is more susceptible to occur. Moreover, removal of the growth restrict mask after growing the ELO III-nitride layer and before growing the III-nitride device layers, such as the p-type layer, makes the unevenness of growth more likely.
(14) In another aspect, the III-nitride layers may be grown at a temperature above 700° C.; in some cases, the growth temperature is over 1000° C. to improve crystal quality. This high growth temperature makes the Sift growth restrict mask decompose into Si atoms and O atoms, which are released into the growth atmosphere. In this condition, the decomposed Si atoms and O atoms during growth of the p-type layer causes the compensation of the p-type layer by Si and O as n-type dopants. Unfortunately, this compensation causes an increase in the series resistance of the p-type layer. The present invention avoids this compensation of the p-type layer by removing the growth restrict mask before the growth of the p-type layer.
(15) Generally, in an emitting device, the III-nitride device layers are grown in the order of: n-type layer, active layer, electron blocking layer (EBL), and p-type layer. Preferably, the growth restrict mask is removed before growing the p-type layer. More preferably, the growth restrict mask is removed after growing the ELO III-nitride layers. By doing this, a smooth surface can be obtained for the island-like III-nitride semiconductor layers after growing the III-nitride device layers.
(16) Thus, there are two advantages to removing the growth restrict mask before the device process. One advantage is to obtain the smooth surface. Another advantage is to avoid the compensation of the p-type layers due to the decomposition of the growth restrict mask. Employing the present invention can solve both issues.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Referring now to the drawings in which like reference numbers represent corresponding parts throughout:
(2) FIGS. **1**(*a*) and **1**(*b*) are schematics of a substrate, growth restrict mask and epitaxial layers, according to embodiments of the present invention.
(3) FIGS. **2**(*a*), **2**(*b*), **2**(*c*), **2**(*d*), **2**(*e*), **2**(*f*), **2**(*g*), **2**(*h*) and **2**(*i*) are schematics and scanning electron microscope (SEM) images comparing results obtain with and without the growth restrict mask, according to embodiments of the present invention.
(4) FIGS. **3**(*a*), **3**(*b*), **3**(*c*), **3**(*d*), **3**(*e*) and **3**(*f*) are scanning electron microscope (SEM) images of island-like III-nitride-based semiconductor layers, according to one embodiment of the present invention.
(5) FIG. **4** is a cross-sectional view of a laser diode device formed from the island-like III-nitride-based semiconductor layers, according to one embodiment of the present invention.
(6) FIGS. **5**(*a*), **5**(*b*), **5**(*c*) and **5**(*d*) illustrate how a dividing support region is formed at periodic lengths along a bar of the device, according to one embodiment of the present invention.
(7) FIGS. **6**(*a*) and **6**(*b*) also illustrate how a dividing support region is formed at periodic lengths

along a bar of the device, according to one embodiment of the present invention.
(8) FIGS. **7**(*a*) and **7**(*b*) show SEM images of the backside of a bar of a device after being removed and the surface of the substrate after removing the bar of the device, for the (1-100) just, (20-21), and (20-2-1) planes, respectively.
(9) FIGS. **8**(*a*), **8**(*b*), **8**(*c*), **8**(*d*), **8**(*e*), **8**(*f*) and **8**(*g*) illustrate a procedure for removing a bar of a device, according to one embodiment of the present invention.
(10) FIGS. **9**(*a*) and **9**(*b*) are SEM images of a bar of a device from different substrate planes.
(11) FIGS. **10**(*a*), **10**(*b*), **10**(*c*), **10**(*d*) and **10**(*e*) are SEM images of a bar of a device from a c-plane (0001) substrate plane.
(12) FIGS. **11**(*a*) and **11**(*b*) illustrate a growth restrict mask used for multiple bars of multiple devices, according to one embodiment of the present invention.
(13) FIGS. **12**(*a*) and **12**(*b*) are SEM images of a very smooth surface for a (1-100) plane without a mis-cut orientation after a bar of a device is removed.
(14) FIGS. **13**(*a*) and **13**(*b*) are SEM images of a surface of a substrate after removing a bar of a device for a (0001) plane with a mis-cut orientation of 0.7 degrees towards the m-plane.
(15) FIG. **14** shows SEM images and material properties for a surface of a (20-21) free standing GaN substrate after removing a bar of a device.
(16) FIGS. **15**(*a*), **15**(*b*), **15**(*c*), **15**(*d*), **15**(*e*) and **15**(*f*) illustrate a procedure for dividing a bar of a device after the bar has been removed from a substrate using a polymer tape, according to one embodiment of the present invention.
(17) FIG. **16** illustrates a coating process for facets of devices, according to one embodiment of the present invention.
(18) FIGS. **17**(*a*), **17**(*b*) and **17**(*c*) illustrate how wire bonds are attached to devices, and a heat sink plate is divided at trenches, according to one embodiment of the present invention.
(19) FIGS. **18**(*a*) and **18**(*b*) illustrate how a heat sink plate is divided to separate devices, according to one embodiment of the present invention.
(20) FIGS. **19**(*a*) and **19**(*b*) illustrate a testing apparatus for devices, according to one embodiment of the present invention.
(21) FIG. **20** illustrates a TO-can package for a laser diode device, according to one embodiment of the present invention.
(22) FIG. **21** illustrates a package for a device, including a heat sink plate, according to one embodiment of the present invention.
(23) FIGS. **22**(*a*) and **22**(*b*) illustrate the etching of layers in layer bending regions, according to one embodiment of the present invention.
(24) FIGS. **23**(*a*) and **23**(*b*) illustrate the structure of a polymer film, according to one embodiment of the present invention.
(25) FIGS. **24**(*a*), **24**(*b*) and **24**(*c*) are schematics illustrating growth with and without the growth restrict mask, according to embodiments of the present invention.
(26) FIGS. **25**(*a*), **25**(*b*), **25**(*c*) and **25**(*d*) are schematics illustrating growth with and without the growth restrict mask, according to embodiments of the present invention.
(27) FIGS. **26**(*a*) and **26**(*b*) are schematics illustrating growth with and without the growth restrict mask, according to embodiments of the present invention.
(28) FIGS. **27**(*a*) and **27**(*b*) are SEM images and schematics illustrating the shape of the remaining space after growing ELO III-nitride layers and III-nitride device layers with and without the growth restrict mask, according to embodiments of the present invention.
(29) FIG. **28** illustrates devices mounted with their top-side down to the heat sink plate, according to one embodiment of the present invention.
(30) FIGS. **29**(*a*) and **29**(*b*) are SEM images illustrating the surface of the III-nitride device layers, according to embodiments of the present invention.
(31) FIGS. **30**(*a*) and **30**(*b*) are SEM images illustrating the connection of III-nitride device layers

after the removal of the growth restrict mask, according to embodiments of the present invention.
(32) FIG. **31** is a flowchart that illustrates a method for dividing a bar of one or more devices.
DETAILED DESCRIPTION OF THE INVENTION
(33) In the following description of the preferred embodiment, reference is made to a specific embodiment in which the invention may be practiced. It is to be understood that other embodiments may be utilized and structural changes may be made without departing from the scope of the present invention.
(34) Device Structure
(35) FIGS. **1**(*a*) and **1**(*b*) are cross-sectional views that illustrate a device structure fabricated according to one embodiment of the present invention.
(36) In the embodiment of FIG. **1**(*a*), a III-nitride-based substrate **101** is provided, such as a bulk GaN substrate **101**, and a growth restrict mask **102** is formed on or above the substrate **101**. Striped opening areas **103** are defined in the growth restrict mask **102**.
(37) No-growth regions **104** result when ELO III-nitride layers **105**, grown from adjacent opening areas **103** in the growth restrict mask **102**, are made not to coalesce on top of the growth restrict mask **102**. Preferably, growth conditions are optimized such that the ELO III-nitride layers **105** have a lateral width of 20 μm on a wing region thereof.
(38) Additional III-nitride device layers **106** are deposited on or above the ELO III-nitride layers **105**, and may include an active region **106***a*, an electron blocking layer (EBL) **106***b*, and a cladding layer **106***c*, as well as other layers.
(39) The thickness of the ELO III-nitride layers **105** is important, because it determines the width of one or more flat surface regions **107** and layer bending regions **108** at the edges thereof adjacent the no-growth regions **104**. The width of the flat surface region **107** is preferably at least 5 and more preferably is 10 μm or more, and most preferably is 20 μm or more.
(40) The ELO III-nitride layers **105** and the additional III-nitride device layers **106** are referred to as island-like III-nitride semiconductor layers **109**, wherein adjacent island-like III-nitride semiconductor layers **109** are separated by the no-growth regions **104**.
(41) The width of the no-growth region **104** can control the amount of the decomposition of the growth restrict mask **102**. The narrower the width of the no-growth region **104**, the lesser the amount of the decomposition of the growth restrict mask **102**. Reducing the amount of the decomposition can alleviate compensation of p-type layers of the III-nitride device layers **106** by the decomposition of the growth restrict mask **102**.
(42) The distance between the island-like III-nitride semiconductor layers **109** adjacent to each other is the width of the no-growth region **104**, which is generally 20 μm or less, and preferably 5 μm or less, but is not limited to these values. Each of the island-like III-nitride semiconductor layers **109** may be processed into a separate device **110**. The device **110**, which may be an LED, LD, Schottky barrier diode, or metal-oxide-semiconductor field-effect-transistor, is processed on the flat surface region **107** and/or the opening areas **103**. Moreover, the shape of the device **110** generally comprises a bar.
(43) In the present invention, the bonding strength between the substrate **101** and the ELO III-nitride layer **105** is weakened by the growth restrict mask **102**. In this case, the bonding area between the substrate **101** and the ELO III-nitride layer **105** is the opening area **103**. The width of the opening area **103** is narrower than the width of island-like III-nitride semiconductor layers **109**. The strength of the bonding between the growth restrict mask **102** and the island-like III-nitride semiconductor layers **109** is weak. Moreover, the island-like III-nitride semiconductor layers **109** bond to the substrate **101** generally only at the opening area **103**. Consequently, the bonding area is reduced by the growth restrict mask **102**, so that this method is preferable for removing the island-like III-nitride semiconductor layers **109**.
(44) Finally, a bottom layer **111** may be present, which may form in the no-growth region **104** between the island-like III-nitride semiconductor layers **109**. If the island-like III-nitride

semiconductor layers **109** are connected to the bottom layer **111**, it may be difficult to remove the island-like semiconductor layers **109** from the substrate **101**.
(45) Device Fabrication
(46) The steps used to fabricate the device **110** using the present invention are described below.
(47) Step 1: Forming a growth restrict mask **102** with a plurality of opening areas **103** directly or indirectly upon a substrate **101**, wherein the substrate **101** is a III-nitride substrate or a hetero-substrate.
(48) Step 2: Growing the ELO III-nitride layers **105** upon the substrate **101** using the growth restrict mask **102**, such that the growth extends in a direction parallel to the striped opening areas **103** of the growth restrict mask **102**, wherein the ELO III-nitride layers **105** do not coalesce.
(49) Step 3: Removing the substrate **101** with the ELO III-nitride layers **105** from the MOCVD reactor. The growth restrict mask **102** is removed from the substrate **101** by wet etching method with an etchant such as hydrofluoride (HF), or buffered HF (BHF).
(50) Step 4: Growing the III-nitride device layers **106**, after the growth restrict mask **102** is removed. The III-nitride device layers **106** are grown on or above the ELO III-nitride layers **105**, and on or above the substrate **101** surface between the ELO III-nitride layers **105**, due to the removal of the growth restrict mask **102**.
(51) Step 5: Fabricating the device **110** at the flat surface region **107** using conventional methods, wherein a ridge structure, p-electrode, pad-electrode etc., are disposed on the island-like III-nitride semiconductor layers **109** at pre-determined positions.
(52) Step 6: Forming a support structure for cleaving at a side facet and a flat surface region **107** of a bar of the device **110**.
(53) Step 7: Removing the bar of the device **110** from the substrate **101**. Step 7.1: Attaching a polymer film to the bar of the device **110**. Step 7.2: Applying pressure to the polymer film and the substrate **101**. Step 7.3: Reducing the temperature of the film and the substrate **101** while the pressure is applied. Step 7.4: Utilizing the different of thermal coefficient between the polymer film and the material of the substrate **101** for removing the bar of the device **110**.
(54) Step 8: Fabricating an n-electrode at a separate area of the device **110**.
(55) Step 9: Breaking the bars into separate devices **110** or chips.
(56) Step 10: Mounting the devices **110** on a heat sink plate.
(57) Step 11: Coating the facets of a laser diode device **110**.
(58) Step 12: Dividing the heat sink plate into separate devices **110**.
(59) Step 13: Screening the devices **110**.
(60) Step 14: Mounting the devices **110** on or into packages.
(61) These steps are explained in more detail below.
(62) Step 1: Forming a Growth Restrict Mask:
(63) A growth restrict mask **102** comprised of patterned SiO.sub.2 is deposited on a substrate **101**, such as an m-plane GaN substrate **101**, In one embodiment, the growth restrict mask **102** is comprised of stripes separated by the opening areas **103**, wherein the stripes have a width of 50 μm, an interval of 50 μm, and are oriented along the <0001> axis. The opening areas **103** are designed with a width of about 2 μm-180 μm, and more preferably, 4 μm-50 μm.
(64) FIG. **1**(*b*) illustrates an alternative embodiment, wherein these techniques are used with a hetero-substrate **101** and a template layer **112**, such as a GaN intermediate layer or underlayer of 2-10 μm, is grown on the hetero-substrate **101**. However, it is not necessary to grow the template layer **112** on the hetero-substrate **101**; instead, the SiO.sub.2 of the growth restrict mask **102** can be formed on the hetero-substrate **101**, and then the ELO III-nitride layers **105** can be grown directly on the growth restrict mask **102** formed on the hetero-substrate **101**.
(65) Step 2: Growing the ELO III-Nitride Layers Upon the Substrate
(66) As shown in FIGS. **1**(*a*) and **1**(*b*), the ELO III-nitride layers **105** are grown on the growth restrict mask **102**. Preferably, the ELO III-nitride layers **105** do not coalesce on top of the growth

restrict mask **102**.

(67) In one embodiment, MOCVD is used for the epitaxial growth of the ELO III-nitride layer **105**. The ELO III-nitride layers **105** are preferably GaN or AlGaN layers in order to obtain a smooth surface. Trimethylgallium (TMGa) and/or triethylaluminium (TMAl) are used as the III elements source; ammonia (NH.sub.3) is used as the raw gas to supply nitrogen; and hydrogen (H.sub.2) and nitrogen (N.sub.2) are used as a carrier gas of the III elements sources. It is important to include hydrogen in the carrier gas to obtain a smooth surface for the epi-layer. The thickness of the ELO III-nitride layer **105** is about 3 μm-100 μm.

(68) Step 3: The Substrate with the ELO III-Nitride Layers is Removed the MOCVD Reactor

(69) The substrate **101** is removed from the MOCVD reactor to remove the growth restrict mask **102** after the ELO III-nitride layers **105** are grown. The growth restrict mask **102** is removed by wet etching with HF or BHF, etc. By doing this, the area which was covered the by growth restrict mask **102** can be used to grow the III-nitride device layers **106**. This increases the growth area on the substrate **101** as compared to before removing the growth restrict mask **102**.

(70) Step 4: Growing a Plurality of III-Nitride Device Layers

(71) After removal of the growth restrict mask **102**, the substrate **101** is loaded back into the MOCVD reactor, for epitaxial growth of the III-nitride device layers **106**.

(72) Tritnethylgallium (TMGa), trimethylindium (TMIn) and triethylaluminium (TMAl) are used as the III elements sources; ammonia (NH.sub.3) is used as the raw gas to supply nitrogen; and hydrogen (H.sub.2) and nitrogen (N.sub.2) are used as a carrier gas of the III elements sources. It is important to include hydrogen in the carrier gas to obtain a smooth surface for the epi-layer.

(73) Saline and Bis(cyclopentadienyl)magnesium (Cp.sub.2Mg) are used as n-type and p-type dopants. The pressure setting typically is 50 to 760 Torr. The III-nitride device layers **106** are generally grown at temperature ranges from 700° C. to 1250° C.

(74) For example, the growth parameters include the following: TMG is 12 sccm, NH.sub.3 is 8 slm, carrier gas is 3 slm, SiH.sub.4 is 1.0 sccm, and the V/III ratio is about 7700. These growth conditions are one example; the conditions can be changed and optimized for each of the layers.

(75) Comparing the Results With and Without the Growth Restrict Mask

(76) The III-nitride device layers **106** obtained with and without the growth restrict mask **102** are illustrated in FIGS. **2**(*a*), **2**(*b*), **2**(*c*), **2**(*d*), **2**(*e*), **2**(*f*), **2**(*g*), **2**(*h*) and **2**(*i*). Specifically, these figures illustrate the results obtained by removing at least part of the growth restrict mask **102**.

(77) As shown in FIG. **2**(*a*), if the growth restrict mask **102** remains in place when the III-nitride device layers **106** are grown, the supply gases, such as TMGa, TMIn, TMAl, NH.sub.3, etc., are not consumed at the exposed area **201** of the growth restrict mask **102**. Instead, a large amount of the supply gases **202** are found near the edges of the ELO III-nitride layers **105**. This results in a non-uniformity of the supply gases **202**.

(78) As shown in FIG. **2**(*b*), after the removal of the growth restrict mask **102**, the supply gases **202** are consumed in the area **203** where the growth restrict mask **102** was removed. The removal of the growth restrict mask **102** decreases the amount of the supply gases **202** near the edges of the ELO III-nitride layers **105**. This results in a more uniformity of the supply gases **202**.

(79) In order to confirm the above mechanism, two samples were prepared. The ELO III-nitride layers **105** of the two samples were prepared at the same time. Both samples had a smooth surface. A first sample did not remove the growth restrict mask **102** before growing the III-nitride device layers **106**, while a second sample removed the growth restrict mask **102** before growing the III-nitride device layers **106**. The two samples were co-loaded into the MOCVD chamber.

(80) In the case where the growth restrict mask **102** was not removed, the excess supply gases **202** at the edges of the ELO III-nitride layer **105** caused a non-uniformity in the growth at the surface of the ELO III-nitride layer **105**. As shown in image (**1**) of FIG. **2**(*c*), the region of the edge on the ELO III-nitride layer **105** sometimes became thicker than the center of the layer. This resulted in a rough surface morphology.

(81) In the case where the growth restrict mask **102** was removed, the excess supply gases **202** at the edges of the ELO III-nitride layers **105** was reduced by eliminating the growth restrict mask **102**. Consumption of the supply gases **202** in the area **203** after removal of the growth restrict mask **102** can alleviate the excess supply gases **202** at the edges of the ELO III-nitride layers **105**, This results in obtaining a smooth surface as shown in image (**2**) of FIG. **2**(*c*).
(82) Furthermore, photoluminescence (PL) measurements were also conducted on the two samples. As shown in FIG. **2**(*d*), the surface morphology affected the PL image. The sample with the growth restrict mask **102**, shown in image (**1**) of FIG. **2**(*d*), had a strong fluctuation in the PL image. On the other hand, the sample without the growth restrict mask **102**, shown in image (**2**) of FIG. **2**(*d*), had drastically reduced fluctuations in the PL image. This shows that this technique can result in improved characteristics for the devices **110**.
(83) As noted above, the III-nitride device layers **106** can include various type of layers. One type is a low-temperature growth layer or an In-containing layer, such as an active layer. Another type is an Al-containing layer, such as an AlGaN cladding layer and/or an EBL layer. Yet another type is a p-type layer, such as a p-GaN layer, a p-InGaN contact layer, and so on. These layers are susceptible to being affected by the non-uniformity of the supply gases. When growing these layers, the present invention is very effective.
(84) Limited Area Epitaxy (LAE)
(85) According to US Patent Application Publication No. US2017/0092810A1, a number of pyramidal hillocks were observed on the surface of m-plane films after growing the epi-layers. Furthermore, a wavy surface and depressed portion have appeared on the growth surface, which made the surface roughness worse. This is a very severe problem. When an LD structure is fabricated on the surface. For that reason, it is better to grow the epitaxial layer on a non-polar and semi-polar substrate, which is well known to be difficult.
(86) For example, according to some papers, a smooth surface can be obtained by controlling an off-angle (>1 degree) of the substrate's growth surface, as well as by using an N.sub.2 carrier gas condition. These are very limiting conditions for mass production, however, because of the high production cost. Moreover, GaN substrates have a large fluctuation of off-angles to the origin from their fabrication methods. For example, if the substrate has a large in-plane distribution of off-angles, it has a different surface morphology at these points in the wafer. In this case, the yield is reduced by the large in-plane distribution of the off-angles. Therefore, it is necessary that the technique does not depend on the off-angle in-plane distribution.
(87) The present invention solves these problem as set forth below. 1. The growth area is limited by the growth restrict mask area from the edge of the substrate. 2. The patterned substrate is a non-polar or semi-polar substrate which has off-angle orientations ranging from −16 degrees to +30 degrees from the m-plane towards the c-plane. In addition, the hetero-substrate with the III-nitride-based semiconductor layer has an off-angle orientation ranging from +16 degrees to −30 degrees from the m-plane towards the c-plane. 3. The island-like III-nitride semiconductor layers **109** have a long side which is the perpendicular to an a-axis of the III-nitride-based semiconductor crystal. 4. During MOCVD growth, a hydrogen atmosphere can be used. 5. The island-like III-nitride semiconductor layers **109** do not coalesce with each other.
(88) Using at least #1, #2 and #3 above, bars with a smooth surface are obtained. It is more preferable that every one of #1, #2, #3, #4 and #5 is conducted.
(89) As shown in FIGS. **3**(*a*), **3**(*b*), **3**(*c*), **3**(*d*), **3**(*e*) and **3**(*f*), the present invention can obtain island-like III-nitride semiconductor layers **109** that have a smooth top surface without pyramidal hillocks and depressed portions, with various planes and various off-angles. The results are explained as following.
(90) FIG. **3**(*a*) shows an image of the island-like III-nitride semiconductor layers **109** on an m-plane (1-100) with a −1.0 degree miscut towards the c-axis.
(91) FIG. **3**(*b*) shows images of the island-like III-nitride semiconductor layers **109** with different

mis-cut orientations of the m-plane (1-100), including (1) 0 degrees, (2) −0.45 degrees, (3) −0.6 degrees, and (4) −1.0 degrees.
(92) FIG. **3**(*c*) shows images of the island-like III-nitride semiconductor layers **109** with different off-angles from the m-plane (1-100) towards the +c-plane, including (1) the (10-10) plane at 0 degrees, (2) the (30-31) plane at +10 degrees, and (3) the (20-21) plane at +15 degrees.
(93) FIG. **3**(*d*) shows images of the island-like III-nitride semiconductor layers **109** with different off-angles from the m-plane (1-100) towards the -c-plane (0001), including (1) the (10-10) plane at 0 degrees, (2) the (20-2-1) plane at −15 degrees, and (3) the (10-1-1) plane at −28 degrees.
(94) FIG. **3**(*e*) shows images of the island-like III-nitride semiconductor layers **109** with different mis-cut orientation from the c-plane (0001) towards the m-plane (1-100), including (1) a 0.2 degree mis-cut, and (2) a 0.8 degree mis-cut.
(95) FIG. **3**(*f*) shows images of the island-like III-nitride semiconductor layers **109** with an AlGaN layer on various planes, including (1-100), (20-21), (20-2-1), (1-100). (20-21) and (20-2-1), wherein the Al composition of the AlGaN layer is 3%-5%. As shown in FIG. **3**(*f*), using the AlGaN layer, a smooth surface can be obtained for the island-like III-nitride semiconductor layers **109**.
(96) Those results have been obtained by the following growth conditions.
(97) In one embodiment, the growth pressure ranges from 60 to 760 Torr, although the growth pressure preferably ranges from 100 to 300 Torr to obtain a wide width for the island-like III-nitride semiconductor layers **109**; the growth temperature ranges from 900° C. to 1200° C.; the VIII ratio ranges from 1000-30,000 and more preferably 3000-10000; the TMG is from 2-20 sccm; NH.sub.3 ranges from 3 to 10 slm; and the carrier gas is only hydrogen gas, or both hydrogen and nitrogen gases. To obtain a smooth surface, the growth conditions of each plane needs to be optimized by conventional methods.
(98) After growing for about 2-8 hours, the ELO III-nitride layer **105** had a thickness of about 8-50 μm and a bar width of about 20-150 μm, wherein the bar width comprises the width of the island-like III-nitride semiconductor layers **109**.
(99) This method can obtain a smooth surface for the ELO III-nitride layers **105** using various semipolar and nonpolar plane substrates **101**, as well as polar c-plane substrates **101**. Thus, the present invention can adopt various planes not depending on an off-angle of the substrate **101**.
(100) III-Nitride Device Layers
(101) FIG. **4** is a cross-sectional side view of a III-nitride semiconductor laser diode device **110** fabricated along a direction perpendicular to an optical resonator.
(102) The device **110** is fabricated at the flat surface region **107** by conventional methods, wherein a ridge structure, p-electrode, n-electrode, pads, etc., are disposed on the island-like III-nitride semiconductor layers **109** at pre-determined positions. (The figure does not describe the bending region **108**.)
(103) The laser diode device **110** is comprised of the following III-nitride device layers **106**, laid one on top of another, in the order mentioned, grown on the ELO GaN-based layers **105** deposited on the growth restrict mask **102**: an n-Al.sub.0.06GaN cladding layer **401**, an n-GaN waveguide layer **402**, an InGaN/GaN multiple quantum well (MQW) active layer **403**, an AlGaN EBL layer **404**, a p-GaN waveguide layer **405**, an ITO cladding layer **406**, an SiO.sub.2 current limiting layer **407**, and a p-electrode **408**.
(104) MOCVD is used for the epitaxial growth of the III-nitride device layers **106**. Trimethylgallium (TMGa), triethylgallium (TEG), trimethylindium (TMIn) and triethylaluminium (TMAl) are used as III elements sources; ammonia (NH.sub.3) is used as the raw gas to supply nitrogen; and hydrogen (H.sub.2) and nitrogen (N.sub.2) are used as a carrier gas of the III elements sources. Saline and Bis(cyclopentadienyl)magnesium (Cp.sub.2Mg) are used as n-type and p-type dopants, respectively. The pressure setting typically is 50 to 760 Torr. The III-nitride device layers **106** are generally grown at temperature ranges from 700 to 1250° C. Other growth parameters include the following: TMG is 12 sccm, NH.sub.3 is 8 slm, carrier gas is 3 slm,

SiH.sub.4 is 1.0 sccm, and the V/III ratio is about 7700. These growth conditions are only one example, and the conditions can be changed and optimized for each of above described layers.
(105) The optical resonator is comprised of a ridge stripe structure, wherein the ridge stripe structure is comprised of the ITO cladding layer **406**, the SiO.sub.2 current limiting layer **407**, and the p-electrode **408**. The optical resonator provides optical confinement in a horizontal direction. The width of the ridge stripe structure is on the order of 1.0 to 30 μm and typically is 10 μm.
(106) Conventional methods, such as photo-lithography and dry etching, can be used to fabricate the ridge strip structure. The ridge depth (from the surface to the ridge bottom) is in the p-GaN guide layer **405**. The ridge depth is pre-determined before dry etching is performed, based on simulation or previous experimental data.
(107) In one embodiment, the p-electrode **408** may be comprised of one or more of the following materials: Pd, Ni, Ti, Pt, Mo, W, Ag, Au, etc. For example, the p-electrode **408** may comprise Pd—Ni—Au (with thicknesses of 3-30-300 nm). These materials may be deposited by electron beam evaporation, sputter, thermal heat evaporation, etc. In addition, the p-electrode **408** is typically deposited on the ITO cladding layer **406**.
(108) The Growth Restrict Mask and the III-Nitride Device Layers
(109) There is a concern in the case where the growth restrict mask **102** is removed before the growth of the III-nitride device layers **106**. The concern arises when the island-like III-nitride semiconductor layers **109** connect with the bottom layer **111**. If the two layers **109**, **111** connect to each other, it is difficult to remove the island-like III-nitride semiconductor layers **109**. However, FIGS. **2**(*f*) and **2**(*g*) illustrate a situation where the two layers **109**, **111** do not connect to each other, wherein FIG. **2**(*f*) shows the island-like III-nitride semiconductor layers **109** separated from the bottom layer **111** by a bottom area **204**, and FIG. **2**(*g*) shows the island-like III-nitride semiconductor layers **109** separated from the bottom layer **111** by bottom areas **204***a* and **204***b*. In both instances, the bottom layer **111** does not grow at the edge of the growth restrict mask **102**, or has a very slow growth rate. This result occurs because the edge of the growth restrict mask **102** shadows the bottom area **204** from the supply gases **202** used in the growth of the island-like III-nitride semiconductor layers **109**.
(110) Step 5: Device Processes
(111) After Step 4, the island-like III-nitride semiconductor layers **109** are separated from each other. The present invention can use either a III-nitride substrate **101** or a hetero-substrate **101**, such as sapphire, SiC, LiAlO.sub.2, Si, etc., as long as it enables growth of an ELO III-nitride layer **105** through a growth restrict mask **102**. In the case using a III-nitride substrate **101**, the present invention can obtain high quality ELO III-nitride layers **105**, and avoid bowing or curvature of the substrate **101** during epitaxial growth due to homo-epitaxial growth. As a result, the present invention can also easily obtain devices **110** with reduced defect density, such as dislocation and stacking faults.
(112) Step 6: Forming a Structure for Cleaving at the Flat Surface Region and Side Facets
(113) As shown in FIGS. **5**(*a*), **5**(*b*), **5**(*c*) and **5**(*d*), the aim of this step is to prepare to divide a bar **501** of the device **110** before the bar **501** of the device **110** is removed from the substrate **101**. A dividing support region **502** is formed at periodic lengths, wherein each period is determined by the device **110** length. For example, in the case of a laser diode device **110**, one period is set to be 300-1200 μm.
(114) The dividing support region **502** is a line scribed by a diamond tipped scriber or laser scriber, as shown in FIG. **5**(*a*); or a trench formed by dry-etching such as RIE (Reactive Ion Etching) or ICP (Inductively Coupled Plasma), as shown in FIG. **5**(*b*); but is not limited to those methods. The dividing support region **502** may be formed on both sides of the bar **501** or on one side of the bar **501**. The depth of the dividing support region **502** is preferably 1 μm or more.
(115) Both cases can divide the bar **501** into separate devices **110** at the dividing support region **502**, because the dividing support region **502** is weaker than any other part. The dividing support

region **502** avoids breaking the bar **501** at unintentional positions, so that it can precisely determine the device **110** length.

(116) The dividing support region **502** is created at the flat surface region **107** in a manner that avoids a current injection region **503**, which is in the ridge structure, and the p-electrode **408**, and the layer bending region **108**, although it may encompass at least a portion of the SiO.sub.2 current limiting layer **407**.

(117) As shown in FIGS. **5**(*a*) and **5**(*b*), the dividing support region **502** is formed at a first facet **504**, and optionally, a second facet **505**, which are easy to process because they are flattened areas. A third facet **506** may be avoided.

(118) As shown in FIG. **5**(*c*), it may be preferable that the dividing support region **502** is formed only at the second facet **505**, in which case, it must use the small width of the island-like III-nitride semiconductor layers **109**. In this case, it can divide the bar **501** of the device **110** precisely.

(119) Moreover, as shown in FIG. **5**(*d*), the p-electrode **408**, dielectric layer **407**, and p-pad for wire bonding and so on, can avoid the dividing support region **502**. By doing this, FIG. **6**(*a*) is changed like FIG. **6**(*b*).

(120) As shown in the SEM images of FIG. **7**(*a*), the back side of the bar **501**, after being removed, has two different portions: one is a separate area that is between the two dashed white lines; another is a wing region that is outside of the separate area. The two different portions have different surface morphologies, which might be an opportunity to prevent the cleave line from going straight. Besides, there are some fluctuation within the wing region due to the interaction between the mask and the back-side surface of the island-like III-nitride semiconductor layers **109**. FIG. **7**(*b*) shows SEM images of the surface of the substrate **101** after the bar **501** has been removed.

(121) Therefore, it is preferable that the dividing support region **502** be formed on the first facet **504** and/or second facet **505**, as shown in FIGS. **5**(*a*), **5**(*b*), **5**(*c*) and **5**(*d*). By doing this, the shape of the dividing support region **502** is formed uniformly. It is much preferable that the blade for breaking the bar **501** contact the back-side of the bar **501**. By doing this, the cleaving starts from the dividing support region **502** at the top surface of the bar **501**.

(122) Step 7: Removing the Bar of the Device from the Substrate

(123) This step of removing the bar **501** is explained using FIGS. **8**(*a*), **8**(*b*), **8**(*c*), **8**(*d*), **8**(*e*), **8**(*f*) and **8**(*g*).

(124) Step 7.1 comprises attaching a polymer film **801** to the bar **501** of the device **110**, as shown in FIG. **8**(*a*). In this embodiment, the polymer film **801** is comprised of a base film **802**, an adhesive **803** and a backing film **804**.

(125) Step 7.2 comprises applying pressure **805** to the polymer film **801** and the substrate **101** using plates **806**, as shown in FIG. **8**(*b*). The aim of applying pressure **805** is to put the polymer film **801** in-between the bars **501** of the devices **110**. The polymer film **801** is softer than the bars **501** of the devices **110**, so the polymer layer **801** can easily surround the bars **501** of the devices **110**. Preferably, the polymer film **801** is heated in order to soften it, which makes it easy for the polymer film **801** to cover the bars **501** of the devices **110**.

(126) Step 7.3 comprises reducing the temperature of the polymer film **801** and the substrate **101**, while maintaining the applied pressure **805**. It is not necessary to apply pressure **805** during the changing of the temperature.

(127) Step 7.4 comprises utilizing the differences in thermal coefficients between the polymer film **801** and the substrate **101** for removing the bars **501** of the devices **110**.

(128) As shown in FIG. **8**(*c*), the polymer film **801** shrinks as the temperature decreases. As a result, the bottom of the polymer film **801** is lower than the top of the bars **501** of the devices **110**, as shown in FIG. **8**(*d*).

(129) As shown in FIG. **8**(*c*), the polymer film **801** can apply the pressure **805** in the horizontal direction at side facets of the bars **501** of the devices **110**, exposing cleaving points **807** and tilting

the bars **501** of the devices **110** downward obliquely **808**. This pressure **805** applied from the side facets allows the bars **501** of the devices **110** to be effectively removed from the substrate **101**. During low temperature, the polymer film **801** maintains the applied pressure **805** from the top of the polymer film **801** to the bars **501** of the devices **110**.
(130) Various methods may be used to reduce the temperature. For example, the substrate **101** and the polymer film **801** can be placed into liquid N.sub.2 (for example, at 77° K.) at the same time while applying pressure **805**. The temperature of the substrate **101** and the polymer film **801** can also be controlled with a piezoelectric transducer. Moreover, the plate **806** that applies the pressure **805** to the polymer film **801** can be cooled to a low temperature before and/or during contact with the polymer film **801**. By doing this, the polymer film **801** is cooled and can apply pressure **805** to the bars **501** of the devices **110** due to a large thermal expansion coefficient.
(131) When reducing the temperature, the substrate **101** and the polymer film **801** may be wetted by atmospheric moisture. In this case, the temperature reduction can be conducted in a dry air atmosphere or a dry N.sub.2 atmosphere, which avoids the substrate **101** and the polymer film **801** getting wet.
(132) Thereafter, the temperature increases, for example, to room temperature, and the pressure **805** is no longer applied to the polymer film **801**. At that time, the bars **501** of the devices **110** have already been removed from the substrate **101**, and the polymer film **801** is then separated from the substrate **101**. As shown in FIG. **8**(*e*), when using a polymer film **801**, especially a polymer film **801** having adhesive **803**, the bars **501** of the devices **110** can be removed using the polymer film **801** in an easy and quick manner.
(133) As shown in FIG. **8**(*f*), there may be an occasion having a different height t among the bars **501** of the devices **110**, depending on a growth condition. In this case, the removal method with the polymer film **801** is good at removing the different height bars **501** of the devices **110**, because these films **801** are flexible and soft, as shown in FIG. **8**(*g*).
(134) The Method of Removing the Bar
(135) Utilizing the different thermal expansion coefficients between the polymer film **801** and the semiconductor material of the devices **110**, pressure in the horizontal direction pressure is applied uniformly to the whole substrate **101**. The bars **501** of devices **110** can be removed from the substrate **101** without breaking the bars **501** of the devices **110**. This has been proven by the high yields achieved in removing the bars **501**.
(136) As shown in the SEM images of FIGS. **9**(*a*), **9**(*b*), **10**(*a*), **10**(*b*), **10**(*c*), **10**(*d*) and **10**(*e*), this method can remove the bar **501** of the device **110** from many different substrate **101** planes, such as (1-100) just, (20-21), (20-2-1), and (0001). In these examples, the length of the bar **501** is about 1.2 mm. Moreover, even though the (20-21) and (20-2-1) planes are not cleaved facets of the bulk GaN substrate **101**, this method can remove the bar **501** without breaking the bar **501** in an easy manner. In other words, the advantage of this method is that the bar **501** can be removed from different substrate **101** planes with the same method, because this method does not depend on the substrate **101** plane. More preferably, it can utilize the cleavage of the m-plane on the Gall crystal when removing the bar **501**. In the case where the substrate **101** is not an m-plane substrate **101**, such as (20-21), (20-2-1) or (0001), the surface of the separate area after the removal contained m-plane as part of its surface, and the bar **501** can be removed by the lesser pressure.
(137) The bar **501** of the device **110** is a rectangular shape with long sides and short sides, as shown in FIGS. **11**(*a*) and **11**(*b*). Pressure is applied to the bar **501** of the device **110** having such a shape from a vertical direction and in a horizontal direction against the long side of the bar **501**, as shown in FIG. **8**(*c*). By doing this, an effective impact can be given to the cleaving point **807**, which removes the bars **501** of the devices **110** from the substrate **101**. The growth restrict mask **102** is preferably eliminated from the substrate **101** by wet etching, etc., before attaching the polymer film **801** to the bars **501** of the devices **110**. Eliminating the growth restrict mask **102** makes a space to apply pressure at the cleaving point **807** underneath the bars **501** of the devices **110**, which can tilt

the bars **501** of the devices **110** obliquely downward **808** as shown in FIG. **8**(*c*).
(138) This method using the polymer film **801** can apply the pressure **805** to the bars **501** of the devices **110** uniformly across a large area and in the proper amount. Selecting the kind of polymer film **801** and/or the temperature, and the ramp-up and/or ramp-down rate of the temperature, can control the amount of pressure **805** applied to the bars **501** of the devices **110**. In addition, this invention is not limited by the ramp-up and/or ramp-down rate of the temperature, and a thermosetting resin film may be used to remove the bars **501** of the devices **110** when raising the temperature. Again, this results in a high yield for removing the bars **501**.
(139) In mass production, it is occasionally difficult to remove every bar **501** of a device **110** on a substrate **101**, especially one with a wide area. Sometimes, there is an occasion where some bars **501** of the devices **110** remain on the substrate **101** after the removal of one or more bars **501** of the devices **110**. In the prior art method described above, it is hard to repeat a removal process due to a metal bonding process between a wafer and a support metal.
(140) On the other hand, this removal method of using the polymer film **801** and the substrate **101** with the ELO III-nitride layers **105** can be repeated many times. When some bars **501** of the devices **110** remain on the substrate **101**, repeating this method allows the remaining bars **501** of the devices **110** to be removed from the substrate **101** completely.
(141) Since this removal method does not include a catastrophic process, such as metal bonding, it can be a repeatable process. By repeating this removal method, almost all of the bars **501** of the devices **110** can be removed from the substrate **101**, including substrates **101** such as 2 inch, 4 inch or other wafer sizes.
(142) Cleaving at a Separate Area of the m-Plane
(143) The cleaving method utilizing the m-plane facet is explained here. FIGS. **7**(*a*) and **7**(*b*) show SEM images of the backside of a bar of a device **110** after being removed and the surface of the substrate **101** after removing the bar **501** of the device **110**, for the (1-100) just, (20-21), and (20-2-1) planes, respectively. It can be seen that the backside of the bar **501** for the (1-100) just plane is relatively smooth surface, as shown in FIG. **7**(*a*), as is the surface of the substrate **101** after removing the bar **501** for the (1-100)-plane, as shown in FIG. **7**(*b*), as compared to the other planes.
(144) As shown in the image of FIG. **12**(*a*) and its enlargement in FIG. **12**(*b*), it is possible to obtain a very smooth surface for the (1-100) plane without a mis-cut orientation after the bar **501** of the device **110** is removed, wherein the surface is smooth enough to be used as a facet for a VCSEL. As shown in the image of FIG. **13**(*a*) and its enlargement in FIG. **13**(*b*), the surface of the substrate **101** after removing the bar **501** of the device **110** is pitted for the (0001) plane with a mis-cut orientation of 0.7 degrees towards the m-plane.
(145) As shown in FIG. **7**(*b*), the backside of the bar **501** of the device **110** for the semipolar (20-21) and (20-2-1) planes had periodic concavity and convexity. In these results, the interface at the cleaving point was an m-plane facet. The cleaving interface was measured using a Laser Scanning Confocal Microscope (LSCM).
(146) The images of FIG. **14**, which include measurements by the LSCM, shows the surface of the (20-21) free standing GaN substrate **101** after removing the bar **501** of the device **110**. The surface of the depressed region is tilted 15 degrees from the surface of the (20-21) substrate **101**, wherein the surface of the (20-21) substrate **101** is tilted 15 degrees from the m-plane. Therefore, the surface of the depressed region is an m-plane. The m-plane of bulk GaN is well known as a facet with high cleavage, and utilizing the cleavage of m-plane for removing the bar **501** is very important and useful. The method of removing the bar **501** can effectively utilize the cleavage of the in-plane, even though the semipolar (20-21) substrate **101** does not have the m-plane as a main surface.
(147) As shown in FIG. **14**, it is preferable that the bar **501**, which comprises semipolar III-nitride-based semiconductor layers, has a periodic concavity and convexity shape after being removed from the substrate **101**. By being separated in this way, the bar **501** can avoid excess and non-

uniform pressure. Consequently, the bar **501** can be divided from the substrate **101** without being broken into a short size. It is much preferable that the backside of the bar **501** comprised of semipolar III-nitride-based semiconductor layers has in-plane at least as part of the back surface. The method could be adopted for different semipolar planes in the same way, which is industrially important.
(148) Cleaving at a Separate Area of the c-Plane
(149) The present invention also attempted this removal method on a c-plane Gail substrate **101**, as shown in FIG. **10**(*a*).
(150) As shown in FIGS. **10**(*b*) and **10**(*c*), the growth restrict mask **102** and opening areas **103** are designed as shown in FIG. **11** without a separating region. The length of the long side of the opening area **103** is set to be 15 mm.
(151) The image of FIG. **10**(*d*), which is enlarged in FIG. **10**(*e*), shows the surface of the substrate **101** after removing the bar **501** of the device **110**. The c-plane is one of the cleavage planes in bulk GaN, so the separate area on the back side of the bar **501** was very smooth. This shows that this removing method can also be used with a c-plane GaN substrate **101**.
(152) Avoiding the Connection of the Bottom Layer and the Island-Like Semiconductor Layer In the present invention, the growth restrict mask **102** is removed before growing the III-nitride device layers **106**. The thickness of the ELO III-nitride layer **105** is at least over 4 μm. The height of the ELO III-nitride layer **105** prevents the bottom layer **111** from growing. In addition, the side facet of the ELO III-nitride layer **105** makes a gap between the island-like III-nitride semiconductor layers **109** and the bottom layer **111** at a bottom area **204**, as shown in images (**1**), (**2**), and (**3**) of FIG. **2**(*e*), This area **204** is very low growth rate due to the reduction of the supply gases. FIGS. **2**(*f*), **2**(*g*) and **2**(*h*) illustrate the width of the bottom area **204***a* **204***b* depends on the shape of the side facet of the ELO III-nitride layer **105**. Various shapes of the side facet of the ELO III-nitride layers **105** are shown in FIG. **2**(*i*).
(153) However, as shown in FIGS. **2**(*g*) and **2**(*h*), the III-nitride device layers **106** generally have two types of shapes for the side facets illustrated by **204***a* and **204***b*.
(154) In one example, as illustrated by **204***a*, the edge of the side facet is located outside a bottom edge of the III-nitride device layers **106**. The distance between the bottom edge and the edge of the side facet is the bottom area **204***a*.
(155) In another example, as illustrated by **204***b*, an edge of the side facet is located against the bottom edge of the III-nitride device layers **106**. The distance between the bottom edge and the edge of the side facet is the bottom area **204***b*.
(156) The first example may be preferred, as the additional distance makes a connection between the bottom layer **111** and the island-like III-nitride semiconductor layers **109** more likely to be avoided.
(157) Step 8: Fabricating an n-electrode at the Separate Area of the Device
(158) Steps 8, 9, 10 and 11 are illustrated by FIGS. **15**(*a*), **15**(*b*), **15**(*c*), **15**(*d*), **15**(*e*), **15**(*f*) and **16**.
(159) After removing the bar **501** from the substrate **101**, the bar **501** remains attached to the polymer film **801**, which is shown with the bar **501** positioned in an upside-down manner on the film **801**, as shown in FIG. **15**(*a*).
(160) FIG. **15**(*b*) shows the back side of the bar **501**, both as a schematic and an SEM image, which has a separate area **1501** between the dividing support regions **502**. The separate area **1501** contacts the substrate **101**, or the underlying layer directly, but is not on the growth restrict mask **102**. Cleaving blades **1502** are used at the dividing support regions **502**.
(161) Then, as shown in FIG. **15**(*c*), a metal mask **1503** can be used to dispose an n-electrode **1504** on the back side of the device **110**.
(162) In the case of forming the n-electrode **1504** of back side of the bar **501** after removing the bar **501** from the substrate **101**, the n-electrode **1504** is preferably formed on the separate area **1501**. This separate area **1501** is kept in a good surface condition for the n-electrode **1504** to obtain low

contact resistivity. The present invention keeps this area **1501** clean until removing the island-like III-nitride semiconductor layers **109**.

(163) The n-electrode **1504** also can be disposed on the top surface of the bar **501**, which is the same surface made for a p-electrode **408**.

(164) Typically, the n-electrode **1504** is comprised of the following materials: Ti, Hf, Cr, Al, Mo, W, Au. For example, the n-electrode may be comprised of Ti—Al—Pt—Au (with a thickness of 30-100-30-500 nm), but is not limited to those materials. The deposition of these materials may be performed by electron beam evaporation, sputter, thermal heat evaporation, etc.

(165) Step 9: Breaking the Bar into Separate Devices

(166) After disposing the n-electrode **1504** in Step 8, each bar **501** is divided into a plurality of devices **110**, as shown in FIG. **15**(*d*). The dividing support region **502** helps divide the bar **501** into the devices **110**, as shown in FIG. **15**(*b*). A breaking method can be used, as well as other conventional methods, but it is not limited to these methods. It is preferable that the cleaving blade **1502** contact a side of the bar **501** which is not formed by the dividing support region **502**, at the position of the dividing support region **502**.

(167) As shown in FIG. **15**(*d*), it is possible that multiple bars **501** laterally disposed are both cleaved and broken into separate devices **110** at dividing support regions **502**. Moreover, it is also possible that multiple bars **501** both laterally and longitudinally disposed are cleaved at dividing support regions **502**. Further, the dividing support regions **502** may be disposed on both sides or one side of the bars **501**.

(168) Step 10: Mounting the Device on a Heat Sink Plate

(169) After Step 9, the divided bar **501** is still on the polymer film **801**. In one embodiment, the polymer film **801** is an ultraviolet (UV) light-sensitive dicing tape that is exposed to UV light, which can reduce the adhesive strength of the film **801**, as shown in FIG. **15**(*e*). This makes it easy to remove the devices **110** from the film **801**.

(170) In this step, a heat sink plate **1505** comprised of AlN is prepared. An Au—Sn solder **1506** is disposed on the heat sink plate **1505**, the heat sink plate **1505** is heated over the melting temperature of the solder **1506**, and the devices **110** on the polymer film **801** are bonded to the heat sink plate **1505** using the Au—Sn solder **1506**. The devices **110** can be mounted on the heat sink plate **1505** in two ways: (1) n-electrode **1504** side down or (2) p-electrode **408** side down. FIG. **15**(*f*) shows the devices **110** mounted to the heat sink plate **1505** using the solder **1506** with the n-electrode **1504** side down. Trenches **1507** in the heat sink plate **1505** separate the devices **110**, wherein the trenches **1507** are used to divide the heat sink plate **1505**, as described in more detail below.

(171) Step 11: Coating the Facets of the Device

(172) Step 11 comprises coating the facets **504** of the device **110**. While a laser diode device **110** is lasing, the light in the device **110** that penetrates through the facets **504** of the device **110** to the outside of the device **110** is absorbed by non-radiative recombination centers at the facets **504**, so that the facet temperature increases continuously. Consequently, the temperature increase can lead to catastrophic optical damage (COD) of the facet **504**.

(173) A facet **504** coating can reduce the non-radiative recombination center. Preventing the COD, it is necessary to coat the facet using dielectric layers, such as AlN, AlON, Al.sub.2O.sub.3, SiN, SiON, SiO.sub.2, ZrO.sub.2, TiO.sub.2, Ta.sub.2O.sub.5 and the like. Generally, the coating film is a multilayer structure comprised of the above materials. The structure and thickness of the layers is determined by a predetermined reflectivity.

(174) In the present invention, the bar **501** of device **110** may have been divided in Step 9 to obtain cleaved facets **504** for multiple devices **110**. As a result, the method of coating the facets **504** needs to be performed on multiple devices **110** at the same time, in an easy manner. In one embodiment, the devices **110** are mounted in a horizontally offset manner on the heat sink plate **1505**, e.g., towards one side of the heat sink plate **1505**, as shown in FIG. **15**(*f*). Then, as shown in FIG. **16**, the

devices **110** and the heat sink plate **1505** are placed on a spacer plate **1601**, and a plurality of spacer plates **1601** are stored in a coating holder **1602**.

(175) Note that it is not always necessary to use a spacer plate **1601**, and the heat sink plate **1505** could be used alone. Alternatively, the heat sink plate **1505** could be mounted on another bar or plate, that is then placed on the spacer plate **1601**.

(176) By doing this, the facets **504** of a number of devices **110** can be coated simultaneously. In one embodiment, the facet **504** coating is conducted at least two times once for the front facet **504** of the devices **110** and once for the rear facet **504** of the devices **110**. The length of the heat sink plate **1505** may be dimensioned to be about the cavity length of the laser diode device **110**, which makes it quick and easy to perform the facet **504** coating twice.

(177) Once the spacer plate **1601** is set in the coating holder **1602**, both facets **504** of the devices **110** can be coated without setting the spacer plate **1601** in the coating holder **1602** again. In one embodiment, a first coating is performed on a front facet **504** which emits the laser light, and a second coating is performed on the rear facet **504** which reflects the laser light. The coating holder **1602** is reversed before the second coating in the facility that deposits the coating film. This reduces the lead time of the process substantially.

(178) Step 12: Dividing the Heat Sink Plate

(179) In this step, as shown in FIG. **17**(*a*), wire bonds **1701** and **1702** are attached to the devices **110**, and then the heat sink plate **1505** is divided at the trenches **1507**, for example, between one or more of the devices **110**. FIG. **17**(*b*) is a top view of FIG. **17**(*a*) that shows the relative placements and positions of the devices **110**, trenches **1507** and bonds **1701**, **1702**. FIG. **17**(*c*) shows the use of separated probes **1703** and wire bonds **1704** with the devices **110**.

(180) FIGS. **18**(*a*) and **18**(*b*) further show how the heat sink plate **1505** is divided to separate the devices **110**, which may occur before or after the attachment of the wire bonds **1701**, **1702**. By doing this, it is easy to separate the devices **110** after the coating process has been completed.

(181) Step 13: Screening the Device

(182) This step distinguishes between defective and non-defective devices **110**. First, various characteristics of the devices **110** are checked under a given condition; such as output power, voltage, current, resistivity, FFP (Far Field Pattern), slope-efficiency and the like. At this point, the devices **110** have already been mounted on the heat sink plate **1505**, so it is easy to check these characteristics.

(183) A testing apparatus **1901** is shown in FIGS. **19**(*a*) and **19**(*b*), wherein the p-electrode **408** and the solder **1506**, which has an electrical continuity to the n-electrode **1504**, are contacted by probes **1902**, **1903**. Then, non-defective devices **110** can be selected and screened by an aging test (life time test).

(184) In one embodiment, it is preferable that testing apparatus **1901** comprise a box or other container, so that an aging test may be conducted with the devices **110** sealed in a dry air or nitrogen atmosphere. Moreover, a heat stage **1904** may be sed to maintain the temperature of the devices **110** during the screening test, for example, 60 degrees, 80 degrees and so on. Photodetectors **1905** may be used to measure light output power **1906**, which identifies non-defective devices **110** that have a constant output power, or which identifies defective devices **110**.

(185) In particularly, in the case of a III-nitride-based semiconductor laser diode device **110**, it is known that when a laser diode is oscillated in a moisture-containing atmosphere, it deteriorates. This deterioration is caused by moisture and siloxane in the air, so the III-nitride-based semiconductor laser diode device **110** needs to be sealed in dry air during the aging test.

(186) Consequently, as shown in FIG. **20**, when a III-nitride-based laser diode **2000** is shipped from a manufacturer, the chip **2001** itself is mounted on a stem **2002** and sealed in a dry air atmosphere using a TO-can package **2003**, wherein the package **2003** includes a window **2004** for light emission.

(187) Generally speaking, the screening or ageing tests are conducted before shipping, in order to

screen out defective devices **110**. For example, the screening condition is conducted according to the specifications of the laser diode device HO, such as a high temperature and a high power.

(188) Moreover, an aging test may be conducted with the device **110** mounted on/into the package **2000**, with the package **2000** sealed in dry air and/or dry nitrogen before screening. This fact makes the flexibility of packaging and mounting of the laser device restrictive.

(189) In the prior art, if defective production happens, the defective products are discarded in the whole TO-CAN package **2000**, which is a great loss for production. This makes it difficult to reduce the production costs of laser diode devices **110**. There is a need to detect defective devices **110** at an earlier step.

(190) In the present invention, coating the facets **504** of the device **110** using a heat sink plate **1505**, on which can be mounted a plurality of the devices **110** in a low horizontal position and then, after the coating process, dividing the heat sink plate **1505** and the devices **110** using the trenches **1507**, allows the devices **110**, with the sub-mount of the heat sink plate **1505**, to be checked in the screening test in a dry air or nitrogen atmosphere.

(191) When doing the screening test, the devices **110** already has two contacts, namely the p-electrode **408** and the solder **1506** on the heat sink plate **1505**, or in the case of flip-chip bonding, the n-electrode **1504** and the solder **1506** on the heat sink plate **1505**. Moreover, the present invention can select defective products using the screening test, when the device **110** is only comprised of the chip and the sub-mount. Therefore, in the case of discarding the defective products, the present invention can reduce the loss more than the prior art, which has great value.

(192) In the case of screening of high-power laser diode devices **110**, it may be preferable that the heat sink plate **1505** has two parts of solder **1506** disposed without electrical continuity. One part of solder **1506** is connected to the p-electrode **408** with a wire (not shown), and another part of solder **1506** is connected to the n-electrode **1504** with a wire (not shown). Moreover, it may be preferable that the p-electrode **408** and n-electrode **1504** are connected by two or more wires to the solder parts **1506**, for example, as shown in FIG. **17**(*c*), which shows the p-electrode **408** connected by two or more wires **1704** to the solder **1506**. In this way, the probes **1703** for applying current to the device **110** can avoid contacting the p-electrode **808** (or n-electrode **1504**) directly, which, in the case of screening of a high-power laser diode, is critical. Specifically, the probes **1703** could break the contacted parts, in particular, in the case of applying a high current density.

(193) Step 14: Mounting the Devices on/into the Packages

(194) In this step, as shown in FIG. **21**, the device **110** (including the heat sink plate **1505**) may be mounted in a package **2101** using solder or another metal to bond the device **110** at the bottom of the package **2101**. Pins **2102** of the package **2101** are connected to the device **110** by wires **2103**. By doing this, current from an external power supply can be applied to the device **110**.

(195) This is more preferable than bonding between the package **2101** and the heat sink plate **1505** using a metal, such as Au—Au, Au—In, etc. This method requires a flatness at the surface of package **2101** and at the back side of the heat sink plate **1505**. However, without the solder, this configuration accomplishes a high thermal conductivity and low temperature bonding, which are big advantages for the device process.

(196) Thereafter, a lid **2104** may enclose the package **2101**. Moreover, a phosphor **2105** can be set outside and/or inside the package **2101**, with a window **2106** allow the light emission to exit the package **2101**. By doing this, the package **2101** can be used as a light bulb or a head light of an automobile.

(197) As set forth herein, these processes provide improved methods for obtaining a laser diode device **110**. In addition, once the device **110** is removed from the substrate **101**, the substrate **101** can be recycled a number of times. This accomplishes the goals of eco-friendly production and low-cost modules. These devices **110** may be utilized as lighting devices, such as light bulbs, data storage equipment, optical communications equipment, such as Li-Fi, etc.

(198) It is difficult to package with a plurality of different types of laser devices **110** in one package

2101. However, this method can overcome this issue, due to being able to perform an aging test without the packaging. Therefore, it is easy to mount the different types of devices **110** in one package **2101**.
(199) Fabricating an LED Device
(200) In the case of fabricating an LED device, the same process may be used until Step 5. This discussion explains briefly how to make two types of LEDs. A type 1 LED has two electrodes (a p-electrode and an n-electrode) on one side of the chip, whereas a type 2 LED has an electrode on opposite sides of chip.
(201) First, in the case of the type 1 LED, the p-electrode and n-electrode are formed in Step 5 on the top surface of the device. Then, from Steps 6-10 are the same process and Steps 11-13 are omitted. In Step 14, the removed chips are mounted on packages and heat sink plates. The backside surface of the chips, the package and the heat sink plate, are bonded using an Ag paste.
(202) Second, in the case of the type 2 LED, almost the same process is used until Step 5, where an ITO electrode is formed on the p-GaN contact layer. In this case, the method of dividing the bar is same. Moreover, it is preferable that the layer bending region is eliminated.
(203) Definitions of Terms
(204) III-Nitride-Based Substrate
(205) The III-nitride-based substrate **101** may comprise any type of III-nitride-based substrate, as long as a III-nitride-based substrate **101** enables growth of III-nitride-based semiconductor layers **105**, **106**, **109** through a growth restrict mask **102**, any GaN substrate **101** that is sliced on a {0001}, {11-22}, {1-100}, {20-21}, {20-2-1}, {10-11}, {10-1-1} plane, etc., or other plane, from a bulk GaN and AlN crystal can be used.
(206) Hetero-Substrate
(207) Moreover, the present invention can also use a hetero-substrate **101**. For example, a GaN template **112** or other III-nitride-based semiconductor layer **112** may be grown on a hetero-substrate **101**, such as sapphire, Si, GaAs, SiC, etc., prior to the growth restrict mask **102**. The GaN template **112** or other III-nitride-based semiconductor layer **112** is typically grown on the hetero-substrate **101** to a thickness of about 2-6 μm, and then the growth restrict mask **102** is disposed on the GaN template **112** or other III-nitride-based semiconductor layer **112**
(208) Growth Restrict Mask
(209) The growth restrict mask **102** comprises a dielectric layer, such as SiO.sub.2, SiN, SiON, Al.sub.2O.sub.3, AlN, AlON, MgF, ZrO.sub.2, etc., or a refractory metal or precious metal, such as W, Mo, Ta, Nb, Rh, Ir, Ru, Os, Pt, etc. The growth restrict mask may be a laminate structure selected from the above materials. It may also be a multiple-stacking layer structure chosen from the above materials.
(210) In one embodiment, the thickness of the growth restrict mask is about 0.05-3 μm. The width of the mask is preferably larger than 20 μm, and more preferably, the width is larger than 40 μm. The growth restrict mask is deposited by sputter, electron beam evaporation, plasma-enhanced chemical vaper deposition (PECVD), ion beam deposition (IBD), etc., but is not limited to those methods.
(211) On an m-plane free standing GaN substrate **101**, the growth restrict mask **102** shown in FIGS. **11**(*a*) and **11**(*b*) comprises a plurality of opening areas **103**, which are arranged in a first direction parallel to the 11-20 direction of the substrate **101** and a second direction parallel to the 0001 direction of the substrate **101**, periodically at intervals p**1** and p**2**, respectively, extending in the second direction. The length a of the opening area **103** is, for example, 200 to 35000 μm; the width b is, for example, 2 to 180 μm; the interval p**1** of the opening area **102** is, for example, 20 to 180 μm; and the interval p**2** is, for example, 200 to 35000 μm. The width b of the opening area **103** is typically constant in the second direction, but may be changed in the second direction as necessary.
(212) On a c-plane free standing GaN substrate **101**, the opening areas **103** are arranged in a first

direction parallel to the 11-20 direction of the substrate **101** and a second direction parallel to the 1-100 direction of the substrate **101**.
(213) On a semipolar (20-21) or (20-2-1) GaN substrate **101**, the opening areas **103** are arranged in a direction parallel to [−1014] and [10-14], respectively.
(214) Alternatively, a hetero-substrate **101** can be used. When a c-plane GaN template **112** is grown on a c-plane sapphire substrate **101**, the opening area **103** is in the same direction as the c-plane free-standing GaN substrate; when an m-plane GaN template **112** is grown on an m-plane sapphire substrate **101**, the opening area **103** is same direction as the m-plane free-standing GaN substrate **101**. By doing this, an m-plane cleaving plane can be used for dividing the bar **501** of the device **110** with the c-plane GaN template **112**, and a c-plane cleaving plane can be used for dividing the bar **501** of the device **110** with the m-plane GaN template **112**; which is much preferable.
(215) III-Nitride-Based Semiconductor Layers
(216) The ELO III-nitride layers **105**, the III-nitride device layers **106** and the island-like III-nitride semiconductor layers **109** can include In, Al and/or B, as well as other impurities, such as Mg, Si, Zn, O, C, H, etc.
(217) The III-nitride device layers **106** generally comprise more than two layers, including at least one layer among an n-type layer, an undoped layer and a p-type layer. The III-nitride device layers **106** specifically comprise a GaN layer, an AlGaN layer, an AlGaInN layer, an InGaN layer, etc. In the case where the device has a plurality of III-nitride-based semiconductor layers, the distance between the island-like III-nitride semiconductor layers **109** adjacent to each other is generally 30 μm or less, and preferably 10 μm or less, but is not limited to these figures. In the semiconductor device, a number of electrodes according to the types of the semiconductor device are disposed at predetermined positions.
(218) Merits of Epitaxial Lateral Overgrowth
(219) The crystallinity of the island-like III-nitride semiconductor layers **109** grown using epitaxial lateral overgrowth (ELO) upon the growth restrict mask **102** from a striped opening are **103** of the growth restrict mask **102** is very high.
(220) Furthermore, two advantages may be obtained using a III-nitride-based substrate **101**. One advantage is that a high-quality island-like III-nitride semiconductor layer **109** can be obtained, such as with a very low defects density, as compared to using a sapphire substrate **101**.
(221) Another advantage in using a similar or the same material for both the epilayers **105**, **106**, **109** and the substrate **101**, is that it can reduce strain in the epilayers **105**, **106**, **109**. Also, thanks to a similar or the same thermal expansion, the method can reduce the amount of bending of the substrate **101** during epitaxial growth. The effect, as above, is that the production yield can be high, in order to improve the uniformity of temperature.
(222) The use of a hetero-substrate **101**, such as sapphire (m-plane, c-plane), LiAlO.sub.2, SiC, Si, etc., for the growth of the epilayers **105**, **106**, **109** is that these substrates **101** are low-cost substrates. This is an important advantage for mass production.
(223) When it comes to the quality of the device **110**, the use of a free standing based substrate **101** is more preferable, due to the above reasons. On the other hand, the use of a hetero-substrate **101** makes it easy to remove the III-nitride-based semiconductor layers **105**, **106**, **109**, due to a weaker bonding strength at the cleaving point **807**.
(224) Also, when a plurality of island-like III-nitride semiconductor layers **109** are grown, these layers **109** are separated from each other, that is, are formed in isolation, so tensile stress or compressive stress generated in each of the island-like III-nitride semiconductor layers **109** is limited within the layers **109**, and the effect of the tensile stress or compressive stress does not fall upon other III-nitride-based semiconductor layers.
(225) Also, as the growth restrict mask **102** and the ELO III-nitride layers **105** are not bonded chemically, the stress in the ELO III-nitride layers **105** can be relaxed by a slide caused at the interface between the growth restrict mask **102** and the ELO III-nitride layers **105**.

(226) Also, the existence of gaps between each the island-like III-nitride semiconductor layers **109**, as shown by no-growth region **104**, results in the substrate **101** having rows of a plurality of island-like III-nitride semiconductor layers **109**, which provides flexibility, and the substrate **101** is easily deformed when external force is applied and can be bent.
(227) Therefore, even if a slight warpage, curvature, or deformation occurs in the substrate **101**, this can be easily corrected by a small external force, which avoids the occurrence of cracks. As a result, the handling of substrates **101** by vacuum chucking is possible, which makes the manufacturing process of the semiconductor devices **110** more easily carried out.
(228) As explained, island-like III-nitride semiconductor layers **109** made of high quality semiconductor crystal can be grown by suppressing the curvature of the substrate **101**, and further, even when the III-nitride-based semiconductor layers **105**, **106**, **109** are very thick, occurrences of cracks, etc., can be suppressed, and thereby a large area semiconductor device **110** can be easily realized.
(229) Flat Surface Region
(230) The flat surface region **107** is between the layer bending regions **108**. Furthermore, the flat surface region **107** is on the growth restrict mask **102** region.
(231) Fabrication of the semiconductor device is mainly performed on the flat surface region **107**. The width of the flat surface region **107** is preferably at least 5 μm, and more preferably is 10 μm or more. The flat surface region **107** has a high uniformity of thickness for each of the semiconductor layers in the flat surface region **107**.
(232) It is not a problem if the fabrication of the semiconductor device is partly formed on the layer bending region **108**. More preferably, the layer at the bending layer region **108** is removed by etching. For example, it is better that at least a part of active layer in the layer bending region **108** is removed using an etching process, such as dry etching or wet etching.
(233) In the case where the semiconductor device is comprised of the island-like III-nitride semiconductor layers **109**, the distance between the island-like III-nitride semiconductor layers **109** adjacent to each other is generally 20 μm or less, and preferably 5 μm or less, but is not limited to these values. The distance between the island-like III-nitride semiconductor layers **109** is the width of a no growth region **104**.
(234) Layer Bending Region
(235) FIGS. **22**(*a*) and **22**(*b*) illustrate how a bended active region **2201** may remain in the device **110**. The definition of a layer bending region **108** is the region outside of the bended active region **2201** including the bended active region **2201**.
(236) If a non-polar or semi-polar substrate is used, the island-like III-nitride semiconductor layers **109** have two or three facets at one side of island-like III-nitride semiconductor layers **109**. The first facet is the main area for forming a ridge structure, while the second and third facets include the layer bending region **108**.
(237) If the layer bending region **108** that includes an active layer remains in the LED chip, a portion of the emitted light from the active layer is reabsorbed. As a result, it is preferable to remove at least a part of the active layer in the layer bending region **108** by etching.
(238) If the layer bending region **108** that includes an active layer remains in the LD chip, the laser mode may be affected by the layer bending region **108** due to a low refractive index (e.g., an InGaN layer). As a result, it is preferable to remove at least a part of the active layer in the layer bending region **108** by etching. Two etchings may be performed, wherein a first etching removes the active layer in the second facet region before removing epi-layers from the substrate **101**, and a second etching removes the active layers in the third facet region after removing epi-layers from the substrate **101**.
(239) The emitting region is a current injection region. In the case of a laser diode, the emitting region is a ridge structure. In the case of an LED, the emitting region is the region for forming a p-contact electrode.

(240) For both the LD and LED, the edge of the emitting region should be at least 1 μm or more from the edge of the layer bending region, and more preferably 5 μm.
(241) From another point of view, an epitaxial layer of the flat area **107** except for the opening area **103** has a lesser defect density than an epitaxial layer of the opening area **103**. Therefore, it is more preferable that the ridge stripe structure should be formed in the flat area **107** including on a wing region.
(242) Semiconductor Device
(243) The semiconductor device **110** is, for example, a light-emitting diode, a laser diode, a Schottky diode, a photodiode, a transistor, etc., but is not limited to these devices. This invention is particularly useful for micro-LEDs and laser diodes, such as edge-emitting lasers and vertical cavity surface-emitting lasers (VCSELs). This invention is especially useful for a semiconductor laser which has cleaved facets.
(244) Polymer Film
(245) The polymer film **801** is used in order to remove the island-like III-nitride semiconductor layers **109** from the III-nitride-based substrate **101** or the GaN template **112** used with the hetero-substrate **101**. In the present invention, dicing tape, including UV-sensitive dicing tape, which are commercially sold, can be used as the polymer film **801**. For example, the structure of the polymer film **801** may comprise triple layers **802**, **803**, **804** or double layers **803**, **804**, as shown in in FIGS. **23**(*a*) and **23**(*b*), but is not limited to those examples. The base film **802** material, for example, having a thickness of about 80 μm, may be made of polyvinyl chloride (PVC). The backing film **804** material, for example, having a thickness of about 38 μm, may be made of polyethylene terephthalate (P.E.T.). The adhesive layer **803**, for example, having a thickness of about 15 μm, may be made of acrylic UV-sensitive adhesive.
(246) When the polymer film **801** is a UV-sensitive dicing tape and is exposed to UV light, the stickiness of the film **801** is drastically reduced. After removing the island-like III-nitride semiconductor layers **109** from the substrate **101**, the polymer film **801** is exposed by the UV light, which makes it is easy to remove.
(247) Heat Sink Plate
(248) As noted above, the removed bar **501** may be transferred to a heat sink plate **1505**, Which may be AlN, SiC, Si, Cu, CuW, and the like. As shown in FIG. **15**(*e*), the solder **1506** for bonding, which may be Au—Sn, Su—Ag—Cu, Ag paste, and the like, is disposed on the heat sink plate **1505**. Then, an n-electrode **1504** or p-electrode **408** is bonded to the solder **1506**. The devices **110** can also be flip-chip bonded to the heat sink plate **1505**.
(249) In the case of bonding LED devices **110** to the heat sink plate **1505**, the size of the heat sink plate **1505** does not matter, and it can be designed as desired.
(250) In the case of bonding laser diode devices **110** to the heat sink plate **1505**, it is preferable that the length of the heat sink plate **1505** be the same or shorter than the length of the laser diode devices **110** for the facet **504** coating process, wherein the length of the laser diode devices **110** is almost the same as the length of the laser cavity. By doing this, it is easy to coat both facets **504** of the laser cavity. If the length of the heat sink plate **1505** is longer than laser cavity, then the heat sink plate **1505** may prevent uniform coating of the laser facets **504**.
(251) Long Width Heat Sink Plate
(252) A long width for the heat sink plate **1505** makes the process of fabricating the laser device **110** more productive. As shown in FIG. **16**, the heat sink plate **1505** is placed on a spacer plate **1601**, and then both are stacked with other heat sink plates **1505** and spacer plates **1601** in the coating holder **1602** for coating a plurality of the devices **110** at the same time. Consequently, a single coating process can coat a large number of devices **110**.
(253) Heat Sink Plate with Trenches
(254) It is preferable that the heat sink plate **1505** have trenches **1507** for dividing the devices **110** as shown in FIGS. **15**(*e*) and **15**(*f*). This structure is useful after the facet **504** coating process,

where the heat sink plate **1505** is divided into one or more devices **110**, for example, single devices **110** or an array of devices **110**. After dividing the heat sink plate **1505**, the devices **110** can be fabricated into packages or modules, such as lighting modules. The trenches **1507** in the heat sink plate **1505** guide the division into the devices **110**. The trenches **1507** can be formed by a wet etching method and mechanically processed before the device **110** is mounted on the heat sink plate **1505**. For example, if the heat sink plate **1505** is made of silicon, wet etching can be used to form the trenches **1507**. Using the trenches **1507** in this manner, reduces the lead time of the process.
(255) Heat Sink Plate with Solder
(256) It is preferable that the length of the solder **1506** be shorter than the device **110** length on the heat sink plate **1505**, as shown in FIG. **15**(*f*). This prevents any wraparound of the solder **1506** to the facets **504**, Which could cause a deterioration of the device **110** characteristics. In particular, wraparound should be avoided for flip-chip mounting.
(257) As shown in FIG. **17**(*b*), after the coating process, the heat sink plate **1505** has wraparound areas, which are the areas surrounded by the dashed lines. The wraparound area has a width W of about 10-20 μm. The coating film will have coated these areas. It is also difficult to avoid coating the solder **1506** with the coating film. Generally, the coating film is selected from one or more dielectric materials, which is why this area does not have conductivity. This is a problem for both conductivity and adhesiveness when a wire **1702** is bonded to the solder **1506**. Therefore, it is preferable that the wire **1702** be placed to avoid the wraparound area. At least, the place of wire bond **1702** should be about 25 μm away from the edge of the heat sink plate **1505**.
ALTERNATIVE EMBODIMENTS
First Embodiment
(258) An III-nitride-based semiconductor device and a method for manufacturing thereof, according to a first embodiment is illustrated in FIGS. **24**(*a*), **24**(*b*) and **24**(*c*).
(259) In the first embodiment, a base substrate **101** is first provided, and a growth restrict mask **102** that has a plurality of striped opening areas **103** is formed on the substrate **101**. In this embodiment, the base substrate **101** is an m-plane substrate made of III-nitride-based semiconductor, which has a mis-cut orientation toward c-axis with −1.0 degree.
(260) The ELO III-nitride layers **105** are grown on or above the substrate **101** and the growth restrict mask **102**. As shown in FIG. **3**(*a*), and image (**4**) of FIG. **3**(*h*), the ELO III-nitride layers **105** are largely uniform with a very smooth surface.
(261) After the growth of the ELO III-nitride layers **105**, the substrate **101** with the layers **105** is removed from the MOCVD reactor in order to remove the growth restrict mask **102**. The growth restrict mask **102** is removed by a wet etching, using an etchant such as HF, BHF, etc., as shown in FIG. **24**(*b*).
(262) Then, the III-nitride device layers **106** are grown on the substrate **101**, as shown in FIG. **24**(*c*). At this time, the III-nitride device layers **106** are grown on both the ELO III-nitride layers **105** and an exposed portion of the substrate **101** where the growth restrict mask **102** has been removed, resulting in the bottom layer **111**.
(263) After the III-nitride device layers **106** are grown, sometimes there is deterioration of the surface morphology, as shown in image (**1**) of FIG. **2**(*c*). It was generally thought that deviations from optimized growth conditions caused the deterioration of the surface roughness. The III-nitride device layers **106** are comprised many layers which are active layers or waveguide layers, so that it is difficult to control the growth conditions. Notably, the excess supply gases to the side facet of the island-like III-nitride semiconductor layers **109** make this deviation from the optimized conditions worse. The optimization of the growth condition avoiding the deterioration of the surface roughness is difficult due to the gases from the side facet.
(264) Therefore, in the present invention, to decrease the effect of the gases at the side facets of the ELO III-nitride layers **105**, the growth restrict mask **102** is removed before growing the III-nitride device layers **106**, at least before the active layer. By doing this, the gases at the side facets of the

ELO III-nitride layers **105** are consumed at the exposed area of the substrate **101** where the growth restrict mask **102** has been removed before reaching the side facets of the ELO III-nitride layers **105**, which avoids supplying excess gases to the side facets.

(265) By doing this, in-plane distribution in terms of the p-type layer thickness improves, which can increase a yield of the mass production. For example, the fluctuation of the p-type layer affects the characteristics of the resulting laser diode. Generally, when forming the laser diode's ridge structure, part of the p-type layer is etched until above the active layer by dry etching method. If there is a fluctuation of the p-type layer's thickness, the remaining p-type layer's thickness after the dry etching also has a fluctuation. This affects the characteristics of the laser diode. Reducing the fluctuations of the p-type layer's thickness is a very important to improve the yields in mass production.

(266) Moreover, removing the growth restrict mask **102** until finishing the growth of the p-type layer at least is also preferable in that it avoids incorporation of decomposed atoms, such as Silicon and Oxygen, from the growth restrict mask **102**, which compensates the dopant of the p-type layer.

(267) Thereafter, the island-like III-nitride semiconductor layers **109** can be processed by the remaining ones of Steps 1-14 as set forth above, in order to obtain a laser diode device.

Second Embodiment

(268) A second embodiment is almost the same as the first embodiment except for the etching of the growth restrict mask **102**. In the first embodiment, the growth restrict mask **102** is totally removed, as shown in FIGS. **24**(*a*) and **24**(*b*). However, the growth restrict mask **102** also can also be at least partly removed, as shown in FIGS. **25**(*a*), **25**(*b*), **25**(*c*) and **25**(*d*), as well as FIGS. **26**(*a*) and **26**(*b*). In both cases, the surface of the island-like III-nitride semiconductor layers **109** is substantially flattened.

(269) As shown in FIGS. **25**(*a*), **25**(*b*) and **25**(*c*), at least a portion of the growth restrict mask **102** is removed by wet etching. Preferably, the portion of the growth restrict mask **102** not covered by the island-like III-nitride semiconductor layers **109** is removed, while the portion of the growth restrict mask **102** covered by the island-like III-nitride semiconductor layers **109** remains. As in the first embodiment, the removal of the growth restrict mask **102** decreases the excess gases supply to the side facets of the island-like III-nitride semiconductor layers **109**. Moreover, the portion of the growth restrict mask **102** remaining under the ELO III-nitride layers **105** is useful when removing the island-like III-nitride semiconductor layers **109** from the substrate **101**. As shown in FIG. **25**(*d*), the remaining portion of the growth restrict mask **102** may be removed by wet etching after growth of the III-nitride device layers **106**.

(270) FIGS. **26**(*a*) and **26**(*b*) show alternative embodiments where at least a portion of the growth restrict mask **102** is removed by wet etching. However, in these examples, additional portions of the growth restrict mask **102** not covered by the island-like III-nitride semiconductor layers **109** remain after etching. As with FIGS. **25**(*a*), **25**(*b*), **25**(*c*) and **25**(*d*), the removal of the growth restrict mask **102** decreases the excess gases supply to the side facets of the island-like III-nitride semiconductor layers **109**. Moreover, the portion of the growth restrict mask **102** remaining under the ELO III-nitride layers **105** is useful when removing the island-like III-nitride semiconductor layers **109** from the substrate **101**.

(271) Before the removal of the growth restrict mask **102**, the shape at the edge of the growth restrict mask **102** is sharp. Thus, the remaining space after the removal of the mask **102** is the same shape. When the III-nitride device layers **106** are grown, the shape of the remaining space is transformed from sharp to round due to the high-temperature conditions when growing in MOCVD, as shown in FIGS. **27**(*a*) and **27**(*b*). Even if the transformation occurred, it is possible to remove the island-like III-nitride semiconductor layers **109**. More preferably, the shape of the remaining space is sharp to determine precisely the breaking point and be easy to remove.

(272) If the growth restrict mask **102** remain, the growth restrict mask **102** prevents the shape at the edge from transforming. As shown in FIG. **25**(*c*), when the III-nitride device layers **106** are grown,

part of the growth restrict mask **102** still remains on the substrate **101**. However, since the mask **101** only remains under the ELO III-nitride layer **105**, the decomposition of the mask **102** is extremely decreased, which reduces the possibility of compensating the p-type layer. The remaining mask **102**, which is under the ELO III-nitride layer **105**, is removed before the removal of the island-like III-nitride semiconductor layers **109**.

Third Embodiment

(273) In a third embodiment, a GaN layer is grown as an ELO III-nitride layer **105** on various off angle substrates **101**. FIGS. **3**(*c*) and **3**(*d*) each include three SEM images with various off angles substrates **101** used with the island-like III-nitride semiconductor layers **109**. The off-angle orientations range from 0 to +15 degrees in FIG. **3**(*c*) and 0 to −28 degrees in FIG. **3**(*d*), from the m-plane towards the c-plane. The present invention can remove the bar from various off angle substrates **101** without breaking the bars, as shown in FIGS. **9**(*a*) and **9**(*b*).

(274) When using a semi-polar substrate **101**, it can obtain the same effect as the first embodiment.

Fourth Embodiment

(275) In a fourth embodiment, a GaN layer is grown as an ELO III-nitride layer **105** on c-plane substrates **101** with two different mis-cut orientation. FIG. **3**(*e*) shows SEM images of two different mis-cut orientation substrates **101** with island-like III-nitride semiconductor layers **109**. The island-like III-nitride semiconductor layers **109** have been removed using the method shown in FIGS. **8**(*a*)-**8**(*e*).

(276) When using a semi-polar substrate **101**, as shown in FIGS. **5**(*a*)-**5**(*d*), it can obtain the same effect as the first embodiment.

Fifth Embodiment

(277) In a fifth embodiment, a sapphire substrate is used as the hetero-substrate **101**. FIG. **1**(*b*) shows the structure of the island-like III-nitride semiconductor layers **109** on the substrate **101**. This structure is almost the same as the first embodiment structure, except for using the sapphire substrate **101** and a buffer layer **112**. The buffer layer **112** is generally used with III-nitride-based semiconductor layers grown on a sapphire substrate **101**. In this embodiment, the buffer layer **112** includes both a nucleation layer and n-GaN layer or undoped GaN layer. The buffer layer **112** is grown at a low temperature of about 500-700° C. degrees. The n-GaN layer or undoped GaN layer is grown at a higher temperature of about 900-1200° C. degrees. The total thickness is about 1-3 mm. Then, the growth restrict mask **102** is disposed on the n-GaN layer or undoped GaN layer. The rest of process to complete the device is the same as the first embodiment.

(278) On the other hand, it is not necessary to use the buffer layer **112**. For example, the growth restrict mask **102** can be disposed on the hetero-substrate **101** directly. After that, the ELO III-nitride layer **105** and/or III-nitride device layers **106** can be grown. In this case, the interface between the hetero-substrate **101** surface and the bottom surface of the ELO III-nitride layer **105** divides easily due to the hetero-interface, which includes a lot of defects.

Sixth Embodiment

(279) This embodiment mounts devices with their top-side down to the heat sink plate **1505**, as shown in FIG. **28**. This embodiment can obtain the bar **501** without an edge-growth, such as shown in the bars of FIG. **2**(*h*).

(280) Image (**1**) of FIG. **2**(*d*) shows the bar with an edge growth having a height. Preferably, the height is 0.3 μm or less. In this invention, an edge growth is defined follow, it locates at the edge of the bar and has a height h over 0.3 μm. The height is defined the difference of the height between the center of the bar **501** and the edge of the bar **501**. By doing this, the top of the bar **501** can be flattened. In a junction-down mounting case, as shown in FIG. **28**, there can be a wider contact area to the heat sink plate **1505** as compared to the bar **501** having edge-growth.

(281) This can improve thermal conductivity. Moreover, it is much preferable that suppressing the edge growth can make a p-type layer thickness uniform.

Seventh Embodiment

(282) In this embodiment, the width of the mask **102** area is changed. In the image shown in FIG. **29**(*a*), the width of the opening area **103** is 50 μm and the width of the mask **102** area is 100 μm. On the other hand, in the image shown in FIG. **29**(*b*), the width of the opening area **103** is 50 μm and the width of the mask **102** area is 50 μm.
(283) In the image of FIG. **29**(*a*), this contrast almost disappears. It can be seen that the contrast to the surface of the bar is a stripe, which is an edge growth.
(284) A wider width of the mask **102** area would increase the supply of gases to the side facet of the bar, which enhances edge growth. The image of FIG. **29**(*b*) is evidence of this edge growth.
(285) Thus, the edge growth can be controlled by designing the width of the mask **102** area. It is much preferable that the width of the mask **102** area 100 μm or less. Using the present invention, it can reduce the height of the edge growth.
Eighth Embodiment
(286) The aim of this embodiment is to avoid a connection between the substrate **101** and the island-like III-nitride semiconductor layers **109** at the edge of a bar **501** after removing the growth restrict mask **102**. When a total thickness of the III-nitride device layers **106** is thicker or the thickness of the growth restrict layer **102** is thinner, a part of the III-nitride device layers **106** can be connected on the substrate **101**, as shown in FIGS. **29**(*a*) and **29**(*b*). The p-layer, which is doped with Mg as a p-type dopant, is apt to grow side facets on the bar.
(287) Whether both parts are connecting or not is dependent on the thickness of the III-nitride device layers **106** after the removal of the growth restrict mask **102**. Therefore, the growth of the III-nitride device layers **106** can be stopped before connecting the two parts, as shown in the image of FIG. **30**(*a*) and its enlargement in FIG. **30**(*b*). It is more preferable to remove the bar **501** in this manner. The thickness of the p-type layer is preferably less than 1 μm.
(288) Process Steps
(289) FIG. **31** is a flowchart that illustrates a method for flattening a surface on the ELO nitride layers **105**, resulting in obtaining a smooth surface with the island-like III-nitride semiconductor layers **109**. The island-like III-nitride semiconductor layers **109** are formed by stopping growth of the ELO III-nitride layers **105** before the ELO III-nitride layers **105** coalesce to each other, and then growing one or more III-nitride device layers **106** on or above the ELO III-nitride layers **106**. The growth restrict mask **102** is removed before growing at least some of the III-nitride device layers **106**, in order to decrease an excess gases supply to side facets of the island-like III-nitride semiconductor layers **109**. The method further comprises preventing compensation of a p-type layer of the III-nitride device layers **106** by decomposed n-type dopants from the growth restrict mask **102**. The steps of the method are described in more detail below.
(290) Block **3101** represents the step of providing a base substrate **101**. In one embodiment, the base substrate **101** is a III-nitride based substrate **101**, such as a GaN-based substrate **101**, or a hetero-substrate **101**, such as a sapphire substrate **101**. This step may also include an optional step of depositing a template layer **112** on or above the substrate **101**, wherein the template layer **11** may comprise a buffer layer or an intermediate layer, such as a GaN underlayer.
(291) Block **3102** represents the step of forming a growth restrict mask **102** on or above the substrate **101**, i.e., on the substrate **101** itself or on the template layer **112**. The growth restrict mask **102** is patterned to include a plurality of striped opening areas **103**.
(292) Block **3103** represents the step of growing one or more III-nitride based layers **105** on or above the growth restrict mask **102** using epitaxial lateral overgrowth (ELO). This step includes stopping the growth of the ELO III-nitride layers **105** before adjacent ones of the ELO III-nitride layers **105** coalesce to each other.
(293) Block **3104** represents the step of growing one or more additional III-nitride device layers **106** on or above the ELO III-nitride layer **105**, thereby forming a bar **501**. These additional III-nitride device layers **106**, along with the ELO III-nitride layer **105**, create one or more of the island-like III-nitride semiconductor layers **109**. Preferably, the III-nitride device layers **106** do not

have an edge growth.
(294) This step may include removing at least part of the growth restrict mask **102** after growing the ELO III-nitride layers **105** and before growing at least some of the III-nitride device layers **106**, in order to obtain a smooth surface for the III-nitride device layers **106**.
(295) The III-nitride device layers **106** may include a low-temperature growth layer, an Indium-containing layer, an Aluminum-containing layer, and/or a p-type layer, and the growth restrict mask **102** may be removed before growing the low-temperature growth layer, Indium-containing layer, Aluminum-containing layer, and/or p-type layer. The growth restrict mask **102** may be removed before growing the p-type layer to avoid compensation of the p-type layer due to decomposition of the growth restrict mask **102**. The growth restrict mask **102** also may be removed before growing at least an active layer of the III-nitride device layers **106**.
(296) The growth restrict mask **102** may be removed before growing the III-nitride device layers **106** to avoid a non-uniformity of supply gases near edges of the ELO III-nitride layers **105**, by decreasing an amount of the supply gases near the edges of the ELO III-nitride layers **105**.
(297) The growth restrict mask **102** may be removed before growing the III-nitride device layers **106** and growth of the III-nitride device layers **106** results in growth of a bottom layer **111** where the growth restrict mask **102** is removed. Preferably, the bottom layer **111** does not connect to the island-like III-nitride semiconductor layers **109**. The bottom layer **111** does not grow, or has a slow growth rate, at an area at an edge of the growth restrict mask **102**, because the edge of the growth restrict mask **102** shadows the area from supply gases. In addition, the ELO III-nitride layers **105** may have a height that prevents the bottom layer **111** from growing or slows the growth of the bottom layer **111**.
(298) A side facet of the ELO III-nitride layers **105** may make a space at a bottom area that reduces the supply gases to the bottom layer **111**, wherein a width of the bottom area depends on a shape of the side facet. An edge of the side facet is located outside a bottom edge line, so that the edge of the side facet reduces the supply gases, which makes a width of the bottom area longer. Alternatively, the side facet does not have an edge located outside a bottom edge line, which makes a width of the bottom area shorter. The supply gases at the side facet are consumed at the growth area before reaching the side facet of the island-like III-nitride semiconductor layers **109**, which avoids supplying the excess gases to the side facet.
(299) Block **3105** represents the step of forming one or more dividing support regions **502** along the bar **501**. The dividing support regions **502** may be formed on a first facet **505** and/or second facet **506** of the bar **501**. In addition, the dividing support regions **502** may be formed on one side or both sides of the bar **501**. The dividing support regions **502** are formed at periodic lengths, wherein each period is determined by the device's length, and each of the dividing support regions **502** comprises a scribed line. In addition, the dividing support regions **502** are created at a flat surface region **107** in a manner that avoids a current injection region **503**.
(300) Block **3106** represents the step of removing devices **110** from the substrate **101**. This step may include applying a polymer film **801** to the bar **501** to remove the bar **501** from the substrate **101** using a cleaving technique on a surface of the substrate **101**, which includes mechanically separating or peeling the island-like III-nitride semiconductor layers **109** from the substrate **101**. The polymer film **801** is applied to the bar **501** by applying pressure to the film **801** and the substrate **101** using plates **806**. The method may also include changing a temperature of the film **801** and the substrate **101**, while the pressure is applied, thereby utilizing a difference in thermal coefficients between the film **801** and the substrate **101** for removing the bar **501** from the substrate **101**. This step may include dividing the bar **501** into one or more devices **110** by cleaving at the dividing support regions **502** formed along the bar **501**. This step may also include the creating of one or more facets **504** on each of the laser diode devices **110**.
(301) Block **3107** represents the step of mounting each of the devices **110** on a heat sink plate **1505** for coating one or more of the facets **504** of the devices **110** created by the cleaving. This step also

includes separating the devices **110** by dividing the heat sink plate **1505** at trenches **1507** in the heat sink plate **1505**. The heat sink plate **1505** may be divided before or after wire bonds **1701**, **1702** are attached to the devices **110**.

(302) Block **3108** represents the resulting product of the method, namely, one or more III-nitride based semiconductor devices **110** fabricated according to this method, as well as a substrate **101** that has been removed from the devices **110** and is available for recycling and reuse.

(303) The device **110** may comprise one or more ELO III-nitride layers **105** grown on or above a growth restrict mask **102** on a substrate **101**, wherein the growth of the ELO III-nitride layers **105** is stopped before adjacent ones of the ELO III-nitride layers **105** coalesce to each other. The device **110** may further comprise one or more III-nitride device layers **106** grown on or above the ELO III-nitride layers **105** and the substrate **101**, wherein at least part of the growth restrict mask **102** is removed after the ELO III-nitride layers **105** are grown and before at least some of the III-nitride device layers **106** are grown, in order to obtain a smooth surface for the III-nitride device layers **106**.

Modifications and Alternatives

(304) A number of modifications and alternatives can be made without departing from the scope of the present invention.

(305) For example, the present invention may be used with III-nitride substrates of various orientations, including c-plane (0001), basal nonpolar m-plane {10-10} families; and semipolar plane families that have at least two nonzero h, i, or k Miller indices and a nonzero 1 Miller index, such as the {20-2-1} planes. Semipolar substrates of (20-2-1) are especially useful, because of the wide area of flattened ELO growth.

(306) In another example, the present invention is described as being used to fabricate different opto-electronic device structures, such as a light-emitting diode (LED), laser diode (LD), Schottky barrier diode (SBD), or metal-oxide-semiconductor field-effect-transistor (MOSFET). The present invention may also be used to fabricate other opto-electronic devices, such as micro-LEDs, vertical cavity surface emitting lasers (VCSELs), edge-emitting laser diodes (EELDs), and solar cells.

CONCLUSION

(307) This concludes the description of the preferred embodiment of the present invention. The foregoing description of one or more embodiments of the invention has been presented for the purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise form disclosed. Many modifications and variations are possible in light of the above teaching. It is intended that the scope of the invention be limited not by this detailed description, but rather by the claims appended hereto.

Claims

1. A method for manufacturing a semiconductor layer, comprising: preparing a plurality of epitaxial lateral overgrowth (ELO) III-nitride layers grown on or above a growth restrict mask on a substrate, wherein adjacent ones of the plurality of ELO III-nitride layers are separated from each other by a no-growth region; removing of the growth restrict mask in the no-growth region, wherein the growth restrict mask does not remain between adjacent ones of the plurality of ELO III-nitride layers and a portion of the growth restrict mask remains between the substrate and the plurality of ELO III-nitride layers; and growing a plurality of III-nitride device layers on or above the plurality of ELO III-nitride layers and the substrate, wherein adjacent ones of the plurality of III-nitride device layers are separated from each other by the no-growth region; wherein: the growth restrict mask is removed after growing the plurality of ELO III-nitride layers and before growing at least some of the plurality of III-nitride device layers; and the plurality of ELO III-nitride layers are grown to a height that prevents growth of a bottom layer in the no-growth region where the growth mask is removed, during the growth of the III-nitride device layers.

2. The method of claim 1, wherein the III-nitride device layers include a p-type layer.
3. The method of claim 1, wherein at least a part of the growth restrict mask is removed where the growth restrict mask is not covered by the ELO III-nitride layers.
4. The method of claim 1, wherein the III-nitride device layers include a low-temperature growth layer, an Indium-containing layer, an Aluminum-containing layer, and/or a p-type layer, and the growth restrict mask is removed before growing the low-temperature growth layer, Indium-containing layer, Aluminum-containing layer, and/or p-type layer.
5. The method of claim 4, wherein the growth restrict mask is removed before growing the p-type layer to avoid compensation of the p-type layer due to decomposition of the growth restrict mask.
6. The method of claim 1, wherein the growth restrict mask is removed before growing at least an active layer of the III-nitride device layers.
7. The method of claim 1, wherein the ELO III-nitride layers and the III-nitride device layers together comprise island-like III-nitride semiconductor layers, and adjacent ones of the island-like III-nitride semiconductor layers are separated from each other by the no-growth region.